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OPTICAL SWITCH UTILIZING OPTICAL BEAM STEERING

Abstract

An optical switch comprising a set of input ports and a set of output ports, each input and output port configured to transport an optical signal having at least one component frequency channel. The switch comprises a first programmable deflection plane configured to deflect beams incident on it from the set of input ports to form a deflected array of beams, and a second programmable deflection plane configured to deflect beams incident on it towards the set of output ports. The switch also comprises a beam steering optical device positioned in the optical path between the first and second programmable deflection planes, configured to remap the deflected array of beams to form a remapped array of beams directed to the second programmable deflection plane, by changing the spatial positioning and/or orientation of each beam from the deflected array of beams in the remapped array of beams such that the spatial positioning and/or orientation of at least one beam of the deflected array of beams is changed differently to at least one other beam of the deflected array of beams.

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Background/Summary

BACKGROUND

[0001] Optical switches are used in optical telecommunication systems to route optical signals through networks. As optical telecommunications systems have become more popular, the quantity of data carried through the networks has increased, putting greater capacity demands on the switches. It is known to use wavelength division multiplexed (WDM) signals to enable each optical fibre in the network to carry multiple data channels, those data channels separated by unique central frequencies and having non-overlapping bandwidths. Wavelength selective switches (WSSs) are used to route WDM signals through the network.

[0002] FIG. 1 illustrates schematically a known $M \times N$ WSS **100**. The $M \times N$ switch comprises N input ports **101** and M output ports **102**. Each port carries multiple data channels. A bank of $1 \times M$ WSSs **103** splits the multiplexed signal from each input port into its separate frequency channels. The demultiplexed data channels are then directed to the M output ports. A bank of $N \times 1$ WSSs **104** at the output combines the data channels into a set of multiplexed signals for output via the output ports **102**. In this way, the $M \times N$ switch is able to redirect any data channel from an input port to any data channel in an output port, subject to the condition that two channels with overlapping frequencies are not routed to the same output port.

[0003] FIG. 2 illustrates schematically a known switch referred to as an add-drop WSS **200**. Add-drop WSS **200** is a special type of WSS in which N input ports **201** are connected to K output ports **202**, where $K > N$. A bank of $1 \times K$ WSSs **203** splits the multiplexed signal from each input port into its separate frequency channels. The demultiplexed data channels are then directed to K space switches **204**. Each space switch **204** can accept data from any of the $1 \times K$ WSSs but can only output data from one of the input ports at a time. The output of each space switch **204** is then output from an output port **202**, otherwise known as a drop port. Space switches are simpler to implement than $N \times 1$ switches, and hence the add-drop WSS of FIG. 2 is preferable to the $M \times N$ WSS of FIG. 1. This is particularly the case when K is much bigger than N , and the data density in the K output ports is much lower than in the N input ports.

[0004] Both the WSSs of FIGS. 1 and 2 are reversible. For example, edge reconfigurable optical add-drop multiplexers (ROADM) are used for transferring optical data between core dense wavelength divisional multiplexing (DWDM) and more coarse wavelength division multiplexing (CWDM). An add-drop WSS of the type in FIG. 2 is used in the “dropping” direction shown in FIG. 2 to transfer from DWDM to CWDM, and in the reverse “adding” direction (from K drop input ports to N output ports) to transfer from CWDM to DWDM.

[0005] The switches shown in FIGS. 1 and 2 are typically implemented by electronic conversion, for example by absorbing light and emitting it again in the desired format. However, electronic conversion has high power requirements and introduces a significant lag in the transmission through the switch. Purely optical methods are preferred because they require much less power than electronic implementations and also enable faster transmission.

[0006] FIG. 3 illustrates a known optical implementation **300** of the $M \times N$ WSS of FIG. 1 or the

add-drop WSS of FIG. 2. N input ports **301** each carry multiple channels. Light through each input port **301** is directed to an imaging telescope **302** where it is focused to a point, and then imaged by a 4F optical system **303**. This produces parallel light beams which are then split into their constituent frequency channels by diffraction grating **304**. The dispersed channels are focussed through lens **305** to a spatial light modulator (SLM) plane **306**, which is typically implemented by a liquid crystal on silicon (LCOS) device. The LCOS device applies a holographic beam deflection to the spectrum of channels incident on it to direct each channel from the N input spectra towards the M spectra on a second SLM plane **308**. The deflected channels pass through a Fourier lens (or lens group) **307** to form the conjugate plane before reaching the second SLM plane **308**. The second SLM plane **308** is typically an LCoS or optical microelectromechanical system (MEMS). The second SLM plane **308** applies an angle deflection to the light incident on it to direct it towards the output ports **309**. The M deflected spectra pass through a lens **310** to generate parallel light beams, which then travel through diffraction grating **311** to recombine the spectra for each of the M output ports **309**. The recombined spectra then pass through 4F optical system **312**. Fourier lens arrangement **314** is used to select the port which the multiplexed signals are then output through. [0007] An undeviated beam **315** is aligned with the central output port **309a**. An angular deflection to this beam channel **316** of a certain magnitude selects an alternative output port **309b**. The deflection of the second SLM plane **308** corrects for the incidence angle from the different spectra on the first SLM plane **306** so that light from any of the input port spectra can be coupled to any of the output ports.

[0008] For the M×N WSS implementation, the second SLM plane **308** of FIG. 3 is a pixelated SLM such as an LCoS. For the add-drop WSS implantation, the second SLM plane **308** of FIG. 3 can be a MEMS array that is pixelated in one direction only (vertical) in order to act as the space switches. In this case, the second SLM plane **308** may be before or after the second diffraction grating **311**.

[0009] These known optical WSSs utilise two separate SLM devices, one for each of the SLM planes **306**, **308**. SLM devices are expensive and limit the compactness of the WSS. Even with the simpler add-drop WSS, the switching is between a vertical column of input spectra on the first SLM plane **306** to a vertical column of space switch areas on the second SLM plane **308**. The large deflection angle on the two SLM planes required to switch light from the top spectra on the first SLM plane **306** to the bottom space switch on the second SLM plane **308** limits the number of accessible output ports M that can be obtained with this arrangement, and hence limits the capacity of a switch having this type of architecture.

[0010] Utilising a single SLM device plane for both SLM planes of an add-drop WSS has been contemplated, for example by U.S. Pat. No. 11,079,551. The single SLM device plane has an area for spectra corresponding to SLM plane **306** of FIG. 3 and an area for space switches corresponding to SLM plane **308** of FIG. 3. FIG. 4 illustrates a single SLM device plane **400** of this type, having a column of space switches **401** and a column of spectra **402**. It can easily be seen that the SLM area is not optimally used. Whilst a large portion of the SLM area is unused, the height of the SLM area limits the number of space switches **401** and the height of the spectra **402**. This in turn limits the number of switchable positions and hence capacity of the system, and also the diffraction efficiency of the system.

[0011] US 2016/0234574 describes utilising a single SLM device for both SLM planes of an add-drop WSS involving switching between a vertical input spectra on the first SLM plane and a horizontal space switch distribution on the second SLM plane. However, the optical implementation of how to achieve this is not disclosed.

[0012] There is a need for a more compact WSS with increased capacity achieved by better utilisation of the available area of its constituent SLM planes so as to maximise the number of ports which can be used.

SUMMARY OF THE INVENTION

[0013] According to an aspect of the invention, there is provided an optical switch comprising: a set of input ports, each input port configured to transport an optical signal having at least one component frequency channel; a set of output ports, each output port configured to transport an optical signal having at least one component frequency channel; a first programmable deflection plane configured to deflect beams incident on it from the set of input ports to form a deflected array of beams; a second programmable deflection plane configured to deflect beams incident on it towards the set of output ports; and a beam steering optical device positioned in the optical path between the first and second programmable deflection planes, the beam steering optical device configured to remap the deflected array of beams from the first programmable deflection plane to form a remapped array of beams directed to the second programmable deflection plane, by changing the spatial positioning and/or orientation of each beam from the deflected array of beams in the remapped array of beams such that the spatial positioning and/or orientation of at least one beam of the deflected array of beams is changed differently to at least one other beam of the deflected array of beams.

[0014] The beam steering optical device may be configured to remap the deflected array of beams by causing them to undergo two individual reflections, refractions or diffractions.

[0015] The beam steering optical device may be configured to remap the deflected array of beams by transposing them in two orthogonal directions.

[0016] The deflected array of beams may have a linear distribution, and the remapped array of beams may have a non-linear distribution.

[0017] The deflected array of beams may be a single column array, and the remapped array of beams may be a two-dimensional array.

[0018] The deflected array of beams may be a two-dimensional array, and the remapped array of beams may be a single column array.

[0019] The deflected array of beams may be a two-dimensional array having a first configuration, and the remapped array of beams may be a two-dimensional array having a different configuration to the first configuration.

[0020] The beam steering optical device may comprise one or more of the following passive components: a prism, a mirror array, a catadioptric system, a freeform lens and mirror, a diffractive optical element, a diffractive optical element array, a holographic optical element and a holographic optical element array.

[0021] The beam steering optical device may comprise one or more of the following electrically switchable components: an optical microelectromechanical system (MEMS) array, and a liquid crystal on silicon (LCoS) device.

[0022] The beam steering optical device may comprise a single optical element only.

[0023] The beam steering optical device may comprise a first optical element, a second optical element and a conjugate Fourier lens, wherein the conjugate Fourier lens is positioned in the optical path between the first and second optical elements.

[0024] The beam steering optical device may comprise a prism comprising: a first retroreflecting structure configured to provide an optical path for a first set of beams of the deflected array of beams; and a second retroreflecting structure configured to provide an optical path for a second set of beams of the deflected array of beams through a periscope prism, so as to alter the conformal arrangement of the first and second sets of beams of the deflected array of beams thereby forming the remapped array of beams having a geometrical distribution which matches the geometrical distribution of the image of the set of output ports on the second programmable deflection plane.

[0025] The beam steering optical device may comprise a mirror array comprising: a first pair of mirrors configured to provide an optical path for a first set of beams of the deflected array of beams; and a second pair of mirrors configured to provide an optical path for a second set of beams of the deflected array of beams, the first and second pairs of mirrors having differently angled surfaces so as to alter the conformal arrangement of the first and second sets of beams of the

deflected array of beams thereby forming the remapped array of beams having a geometrical distribution which matches the geometrical distribution of the image of the set of output ports on the second programmable deflection plane.

[0026] The beam steering optical device may comprise a prism comprising two orthogonal Amici Roof prisms configured to provide an optical path for the deflected array of beams so as to remap the deflected array of beams from a first configuration in a first direction to form the remapped array of beams having the first configuration in a direction orthogonal to the first direction.

[0027] The beam steering optical device may further comprise a retroreflecting mirror at the output of the orthogonal Amici Roof prisms.

[0028] The beam steering optical device may comprise three differently angled reflecting surfaces configured to provide an optical path for the deflected array of beams so as to remap the deflected array of beams from a first configuration in a first direction to form the remapped array of beams having the first configuration in a direction orthogonal to the first direction.

[0029] The beam steering optical device may further comprise: an entrance micro lens located in the optical path input to the beam steering optical device from the first programmable deflection plane; and/or an exit micro lens located in the optical path output from the beam steering optical device to the second programmable deflection plane.

[0030] The optical switch may comprise a single spatial light modulator (SLM) panel comprising both the first programmable deflection plane and the second programmable deflection plane.

[0031] The images of the beams on the first programmable deflection plane and the images of the remapped array of beams on the second programmable deflection plane may cover over 80% of the area of the SLM panel.

[0032] The optical switch may further comprise an anamorphic correction telescope configured to focus the output of each input port of the set of input ports to an anamorphic focus image at an anamorphic focus point; wherein the beam steering optical device comprises: a first structure positioned adjacent to and above the anamorphic focus point, and a second structure positioned adjacent to and below the anamorphic focus point.

[0033] The optical switch may further comprise a lens configured to focus the output of each input port of the set of input ports to an input circular focus image at an input circular focus point; wherein the beam steering optical device comprises: a first structure positioned adjacent to and above the input circular focus point, and a second structure positioned adjacent to and below the input circular focus point.

[0034] The optical switch may further comprise an optical dispersion device configured to spatially separate the component frequency channels of optical signals incident on it from the set of input ports to form a set of dispersed optical signal beams; wherein the first programmable deflection plane is a spectral plane positioned in the optical path of the set of dispersed optical signal beams such that the set of dispersed optical signal beams form an array of images on the first programmable deflection plane, the first programmable deflection plane configured to apply a programmable deflection to each beam individually to form the deflected array of beams.

[0035] The optical switch may further comprise a multiplexer in the optical path between the first programmable deflection plane and the beam steering optical device, the multiplexer configured to multiplex the deflected array of beams from the first programmable deflection plane.

[0036] The first programmable deflection plane may be a space switch plane configured to deflect beams of non-dispersed optical signals to form the deflected array of beams, the optical switch further comprising: an optical dispersion device positioned in the optical path between the beam steering optical device and the second programmable deflection plane, the optical dispersion device configured to spatially separate the component frequency channels of the remapped array of beams, wherein the second programmable deflection plane is a spectral plane configured to apply a programmable deflection to each spatially separated component frequency channel beam of the remapped array of beams to form a spatially separated output array of beams; and a multiplexer in

the optical path between the second programmable deflection plane and the set of output ports, the multiplexer configured to multiplex the spatially separated output array of beams to form a multiplexed output array of beams.

[0037] The multiplexer may be the optical dispersion device.

[0038] Both the first programmable deflection plane and the second programmable deflection plane may be spectral planes configured to deflect each image of a dispersed optical signal individually.

[0039] The optical switch may further comprise an optical dispersion device positioned in the optical path between the set of input ports and the first programmable deflection plane, the optical dispersion device configured to spatially separate the component frequency channels of optical signals from the set of input ports to form a set of dispersed optical signal beams; wherein the set of dispersed optical signal beams form an array of images on the first programmable deflection plane, the first programmable deflection plane configured to apply a programmable deflection to each beam of the array of beams individually to form the deflected array of beams.

[0040] The second programmable deflection plane may be configured to apply a programmable deflection to each spatially separated component frequency channel beam of the remapped array of beams to form a spatially separated output array of beams; the optical switch further comprising a multiplexer in the optical path between the second programmable deflection plane and the set of output ports, the multiplexer configured to multiplex the spatially separated output array of beams to form a multiplexed output array of beams.

[0041] The first programmable deflection plane may be a space switch plane configured to deflect beams of non-dispersed optical signals incident on it to form the deflected array of beams and the second programmable deflection plane may be a space switch plane configured to deflect beams of non-dispersed optical signals incident on it towards the set output ports.

[0042] The optical path between the set of input ports and the set of output ports may not include any optical dispersion devices.

[0043] The first pair of mirrors may comprise a first mirror and a second mirror, the first and second mirrors being parallel to one another; and the second pair of mirrors may comprise a third mirror and a fourth mirror, the third and fourth mirrors being parallel to one another.

[0044] The first set of beams of the deflected array of beams may be incident on the first mirror in a first direction and leave the second mirror in the first direction; and the second set of beams of the deflected array of beams may be incident on the third mirror in a second direction and leave the fourth mirror in the second direction.

[0045] The length of the optical path within the mirror array for the first set of beams of the deflected array of beams may be equal to the length of the optical path within the mirror array for the second set of beams of the deflected array of beams.

[0046] The optical path for the first set of beams and the optical path of the second set of beams may further comprise an optical system located between the first pair of mirrors and between the second pair of mirrors.

[0047] The optical system may comprise a Fourier lens and/or a 4F optical system.

[0048] The optical switch may comprise a mirror optic positioned in the optical path between the set of input ports and the first programmable deflection plane and between the first programmable deflection plane and the beam steering optical device, wherein the mirror optic is configured to direct the deflected light from the first programmable deflection plane towards the beam steering device.

[0049] The optical switch may comprise a mirror optic positioned in the optical path between the beam steering optical device and the second programmable deflection plane and between the second programmable deflection plane and the set of output ports, wherein the mirror optic is configured to direct the remapped array of beams to the second programmable deflection plane.

[0050] The mirror optic may comprise a pair of mirrors separated by a gap.

[0051] The optical path between the set of input ports and the beam steering optical device may

comprise an optical component configured to apply an astigmatism to the beams from the input ports and the optical path between the beam steering device and the set of output ports may comprise an optical component configured to remove the astigmatism from the beams.

[0052] A beam incident on the first mirror may be divided into a first portion aligned with a first plane and a second portion aligned with a second plane which is orthogonal to the first plane; and the beam steering optical device may be configured to focus the first portion on the first mirror and focus the second portion on the second mirror.

[0053] The first plane may be the steering plane and the second plane may be the dispersion plane.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0054] The present invention will now be described by way of example with reference to the accompanying drawings. In the drawings:

[0055] FIG. 1 illustrates a known $M \times N$ WSS;

[0056] FIG. 2 illustrates a known add-drop WSS;

[0057] FIG. 3 illustrates a known implementation of an $M \times N$ or add-drop WSS;

[0058] FIG. 4 illustrates an example distribution of spectra and space switches on a single prior art LCoS device;

[0059] FIG. 5 illustrates an add-drop WSS;

[0060] FIGS. 6a and 6b illustrate the optical path of the add-drop WSS of FIG. 5;

[0061] FIG. 7 illustrates an $M \times N$ WSS;

[0062] FIGS. 8a and 8b illustrate the optical path of a further add-drop WSS;

[0063] FIG. 9 illustrates the add-drop WSS of FIGS. 8a and 8b;

[0064] FIGS. 10a and 10b illustrate the optical path of a further add-drop WSS;

[0065] FIGS. 11, 12 and 13 illustrate add-drop WSSs with redistributed input and output ports;

[0066] FIGS. 14a and 14b illustrate an example distribution of spectra and space switches on a single LCoS device;

[0067] FIG. 15 illustrates a prism array implementation of a beam steering optical device;

[0068] FIG. 16 illustrates a mirror array implementation of a beam steering optical device;

[0069] FIG. 17 illustrates a further prism array implementation of a beam steering optical device;

[0070] FIG. 18 illustrates a further mirror array implementation of a beam steering optical device;

[0071] FIGS. 19a and 19b illustrate the optical path of an add-drop WSS having groups of input ports separated by a gap;

[0072] FIG. 20 illustrates an example distribution of spectra and space switches on a single LCoS device of an add-drop WSS having the optical path of FIGS. 19a and 19b;

[0073] FIGS. 21a and 21b illustrate the optical path of a further add-drop WSS;

[0074] FIG. 22 illustrates an example distribution of spectra and space switches on a single LCoS device of an add-drop WSS having the optical path of FIGS. 21a and 21b;

[0075] FIG. 23 illustrates an add-drop WSS having the optical path of FIGS. 21a and 21b;

[0076] FIG. 24 illustrates an add-drop WSS having the optical path of FIGS. 21a and 21b;

[0077] FIG. 25 illustrates a mirror array implementation of the optical structure;

[0078] FIG. 26 illustrates a prism array implementation of the optical structure;

[0079] FIG. 27 illustrates a further add-drop WSS;

[0080] FIGS. 28a, 28b and 28c illustrate detailed views of two composite structures of the add-drop WSS of FIG. 27;

[0081] FIG. 29 illustrates a further add-drop WSS;

[0082] FIG. 30 illustrates the add-drop WSS of FIG. 29 being operated in reverse;

[0083] FIG. 31 illustrates a further add-drop WSS;

[0084] FIG. **32** illustrates a further example of an $M \times N$ WSS;
[0085] FIGS. **33a**, **33b**, **33c**, **33d** and **33e** illustrate four types of split aperture mirror (SAM);
[0086] FIG. **34** illustrates an arrangement of multiple WSSs of the type seen in FIG. **32**.
[0087] FIG. **35** illustrates a further mirror array implementation of a beam steering optical device;
[0088] FIGS. **36a**, **36b** and **36c** illustrate further mirror array implementations of the beam steering optical device;
[0089] FIGS. **37a**, **37b** and **37c** illustrate further mirror array implementations of the optical structure; and
[0090] FIG. **38** illustrates a further implementation of the optical structure.

DETAILED DESCRIPTION

[0091] The following describes several exemplary optical switches which utilise two programmable deflection planes and a beam steering optical device to optically route light from a set of input ports to a set of output ports. The described optical switches may be WSSs. These WSSs may be implemented as add-drop WSSs (adWSS), for example for transferring light from core DWDM networks to lower capacity CWDM networks. Alternatively, the described WSSs may be implemented as $M \times N$ WSSs. All the examples described herein use optical components to route the light through the switches. There is no absorption and re-emission of light.

[0092] FIG. **5** illustrates an exemplary add-drop WSS **500** in which a single panel **507** comprises two programmable deflection planes **508**, **514**. This single panel may, for example, be an SLM device. FIGS. **6a** and **6b** illustrate the detailed optical path of the light as it is routed through the components of the switch. FIG. **6a** illustrates the path in the switching plane, i.e. the y axis shown is the switching axis and the z axis is the optical axis. FIG. **6b** illustrates the optical path in the dispersion plane, i.e. the x axis shown is the dispersion axis and the z axis is the optical axis. Thus, FIG. **6b** illustrates an orthogonal plane to FIG. **6a**.

[0093] In the adWSS of FIG. **5**, light from a set of N ports **501** is routed to a set of K drop ports **502**. Each port may comprise one or more optical fibres. Each port transports an optical signal having at least one component data channel. Typically, each port has a plurality of data channels. Each channel transports data. Each channel has a frequency band which has a different centre frequency to the other channels of that port. The channels of a port may have the same or different bandwidths. The bandwidths of the channels of a port are non-overlapping. As will be described below, each channel from each input port **501** may be routed to any individual output port **502**.

[0094] The optical signals from the input ports **501** pass through coupling lenses **503** which form an optical beam of the N ports before a 4F imaging system. There is a coupling lens for each input port. The 4F imaging system comprises two lenses **504** and **506** separated by a diffraction grating **505a** at the Fourier plane of lens **504**. The diffraction grating **505a** spatially separates the component frequency channels of the optical signal beams incident on it to form a set of dispersed optical signal beams. In other words, each optical signal from an input port is spread into its frequency spectrum by the diffraction grating **505a**. Thus, the diffraction grating **505a** acts to demultiplex each optical signal into its component data channels.

[0095] A first programmable deflection plane **508** is positioned in the optical path of the set of dispersed optical signals such that the set of dispersed optical signal beams form spectra **517** on the first programmable deflection plane **508**. In the example of FIG. **5**, three input ports **501** are illustrated, each of which forms a spectrum **518** on the first programmable deflection plane **508**. Each spectrum forms an array of beams of the component data channels of the input port which are imaged on the first programmable deflection plane **508**, each beam being a data channel. The first programmable deflection plane **508** applies a programmable deflection to each beam of the array of beams individually to form a deflected array of beams.

[0096] The degree of deflection applied to each beam is so as to change the angle of that beam to direct its data channel on a path towards a selected one of the output ports **502**. The deflection applied by the first programmable deflection plane **508** is reconfigurable. Thus, a controller

providing a control signal to the first programmable deflection plane **508** may change the deflection applied by the first programmable deflection plane **508** to each individual beam of the spectra from each input port, so as to output the desired channels to the desired ports.

[0097] The deflected array of beams from the first programmable deflection plane **508** then travels through a lens **509** and diffraction grating **505b** in the Fourier plane of lens **509**. The diffraction grating **505b** may be the same as or different to the diffraction grating **505a**. The diffraction grating **505b** recombines the deflected spectra for each output port **502**. Thus, the diffraction grating **505b** acts to multiplex the spectra. An imaging lens **510** images the optical signal from the diffraction grating **505b** onto a beam steering optical device **511**.

[0098] The beam steering optical device **511** remaps the beams incident on it to form a remapped array of beams which it directs to a second programmable deflection plane **514**. The optical device **511** is positioned in the optical path between the first and second programmable deflection planes. It reverses the direction of the light of the beam incident on it. It also shifts the beam path incident on it so that the alignment of the beam path with the space switches on the second programmable deflection plane is optimised. It does this by independently controlling the spatial positioning and/or orientation of each beam from the deflected array of beams from the first programmable deflection plane in the remapped array of beams. In this way, it changes the spatial positioning and/or orientation of at least one beam from the deflected array of beams differently to at least one other beam of the deflected array of beams. The optical device may be a mirror, mirror array, freeform optic or a retroreflecting prism. Exemplary optical implementations of the optical device **511** will be described in detail below.

[0099] The remapped array of beams is then directed through two lenses **512** and **513**. Lenses **512** and **513** are in a telescope arrangement, and act to image the light output from the optical device **511** onto the second programmable deflection plane **514**. The telescope arrangement may be so as to magnify the remapped array of beams onto the second programmable deflection plane **514**. Since the spectra have been recombined at the diffraction grating **505b**, the remapped array of beams in FIG. 5 are multiplexed light signals. The second programmable deflection plane **514** acts as a column of vertical space switches **518**, each of which applies a programmable deflection to each beam of the array of beams incident on it. There are the same number of space switches as there are drop ports, in this case K space switches and K drop ports. The deflection corrects for the angle each input port spectrum makes with the space switch so as to efficiently couple each multiplexed light signal to the selected output port. This ensures a directionless operation. The deflection applied by the second programmable deflection plane **514** is reconfigurable. Thus, a controller providing a control signal to the second programmable deflection plane **514** may change the deflection applied by the second programmable deflection plane **514** to each remapped beam from the beam steering optical device **511** so as to output the desired channels to the desired ports.

[0100] The light deflected from the second programmable deflection plane **514** passes through lens **515** creating a single beam at the lens array **516** for coupling to the output ports **502**.

[0101] The number of angular switching positions in the spectrum channels **517** is equal to the number of output ports K. The number of switching positions in the space switches **518** is equal to the number of input ports N.

[0102] Where there are N input ports and K output ports, and $K > N$, N output ports may be used for direct transit from the N input ports. K-N output ports may then be used as drop ports.

[0103] FIGS. 6a and 6b illustrate a more detailed optical path for the configuration shown in FIG. 5. This includes the optical components shown in FIG. 5 and also some further optional optical components. Coupling optics **503** includes a polarisation conversion system **601**. This corrects for uncertain polarisation from the input optical fibres with the coupling optics **503**.

[0104] The polarisation conversion system **601** may comprise a birefringent crystal and a patterned retarder. Coupling optics **516** may also comprise a polarisation conversion system (not shown) for coupling the optical signals to the output ports in a directionless manner. This polarisation

conversion system may comprise a birefringent crystal and a patterned retarder.

[0105] The beams of each of the input ports are focused at input circular focus points **602**. An anamorphic converter **603** is used to convert the circular beam at input circular focus points **602** to an elongated beam focused at input anamorphic focus points **604**. A large beam size improves the efficiency of the hologram zones in the switching axis y on the programmable deflection plane **508**. However, a narrow width in the dispersion axis x enables the spectra to fit onto the programmable deflection plane and also maximises passband performance. Hence the anamorphic converter is used to increase the length of the light beam in the switching axis and reduce the length of the light beam in the dispersion axis, leading to the non-circular elongated shape of the light beam. The anamorphic converter **603** may comprise two anamorphic lenses. Alternatively, the anamorphic converter **603** may comprise four cylindrical lenses. An anamorphic converter **605** is positioned prior to the coupling optics **516** at the output of the switch in order to convert the light beam from an output anamorphic focus point **606** to a circular shape at output circular focus point **607** for output from the output ports. The circular shape is better matched to the shape of the output fibres. In FIGS. **6a** and **6b**, the anamorphic converters **603** and **605** each have an anamorphic lens having different focal lengths in the switching axis and the dispersion axis, and crossed cylindrical lenses. In the switching axis and dispersion axis the two different telescopes have a different magnitude of magnification at the same image plane.

[0106] Optionally, entrance micro-lenses **608a** are located in the optical path input to the beam steering optical device **511**. There is an entrance micro-lens for each input port. Optionally, exit micro-lenses **608b** are located in the optical path output from the beam steering optical device **511**. There is exit micro-lens for each output port. These micro-lenses **608a**, **608b** better transmit the light beam within the beam steering optical device **511**. They are useful if the depth of field of the lens **510** is not significantly greater than the required optical length. Both the entrance and exit micro-lenses may be used. Alternatively, only one of the entrance and exit micro-lenses may be used.

[0107] FIG. **7** illustrates an $M \times N$ WSS **700**. An $M \times N$ WSS has no space switches, but instead a further set of spectra on the second programmable deflection plane. Thus, both the first and second programmable deflection planes are spectral planes configured to deflect each beam of a dispersed optical signal individually. The features of FIG. **7** which are the same as those of FIG. **5** and operate in the same manner are illustrated with the same reference numerals. The detail of those features is as described above with respect to FIGS. **5**, **6a** and **6b**. The structure and operation of the $M \times N$ WSS differs from the description of the adWSS of FIG. **5** above by the following details.

[0108] In the $M \times N$ WSS of FIG. **7**, N input ports **501** are routed to M output ports **701**. The ports each have a plurality of data channels as described with respect to FIG. **5**. Optical signals from the input ports **501** pass through the 4F imaging system provided by lenses **504** and **506**. The optical signals are dispersed by diffraction grating **505a** and form a spectra **517** on the first programmable deflection plane **508**. The first programmable deflection plane deflects the optical signals incident on it as described above. The deflected array of beams is incident on a lens **702**. Lens **702** is positioned so that the spectra **517** and beam steering optical device **511** are in Fourier planes. Lens **702** images the deflected spectra onto the beam steering optical device **511**.

[0109] The beam steering optical device **511** remaps the deflected spectra (which are an array of beams) incident on it to form a remapped spectra (which is a remapped array of beams) which it directs to the second programmable deflection plane **514**. Lens **703** is positioned between the optical device **511** and the second programmable deflection plane **514**. Lens **703** images the remapped spectra from the optical device **511** onto the spectral plane **704** of the second programmable deflection plane **514**.

[0110] The second programmable deflection plane **514** deflects the light incident on it such that it is deviated in a common direction. Thus, the second programmable deflection plane **514** applies a programmable deflection to each spatially separated component frequency channel beam of the

remapped array of beams to form a spatially separated output array of beams. Lenses **705** and **706** are positioned in a 4F arrangement between the second programmable deflection plane **514** and the output ports **701**. A diffraction grating **707** is positioned in the Fourier plane between lenses **705** and **706**. The spatially separated output array of beams deflected by the second programmable deflection plane **514** passes through the lenses **705** and **706** and diffraction grating **707**. Diffraction grating **707** recombines the spatially separated output array of beams for output from the switch. Thus, the diffraction grating **707** acts to multiplex the spectra to form a multiplexed array of beams for output from the switch **700**. The multiplexed signals are beamed to the lens array **516** by the lenses **705** and **706**. The lens array **516** couples the multiplexed signals to the output ports **701**.

[0111] The first and second programmable planes may be implemented on an SLM device. Failure of that SLM device during operation may occur. If this happens and the SLM device no longer applies the programmed deflection, then a default mode may be operated in which the SLM device sends the data from the input ports to specific output ports in a “straight through” operation.

[0112] FIGS. **8a** and **8b** illustrate the detailed optical path of a further example of an add-drop WSS **800**. FIG. **8a** illustrates the path in the switching plane, i.e. the y axis shown is the switching axis and the z axis is the optical axis. FIG. **8b** illustrates the optical path in the dispersion plane, i.e. the x axis shown is the dispersion axis and the z axis is the optical axis. Thus, FIG. **8b** illustrates an orthogonal plane to FIG. **8a**. FIG. **9** illustrates an implementation **900** of the arrangement of FIGS. **8a** and **8b**. The features of FIGS. **8a**, **8b** and **9** which are the same as those of FIGS. **5**, **6a** and **6b** and operate in the same manner are illustrated with the same reference numerals. The detail of those features is as described above with respect to FIGS. **5**, **6a** and **6b**. The structure and operation of the adWSS of FIGS. **8a**, **8b** and **9** differs from the description of the adWSS of FIGS. **5**, **6a** and **6b** above by the following details.

[0113] The adWSS of FIGS. **8a**, **8b** and **9** differs from that of FIGS. **5**, **6a** and **6b** by the use of a different telecentric 4F system for projection. This enables the beam steering optical device **511** to be positioned in the same place as the input anamorphic focus point **804** or the input circular focus point **803**.

[0114] N input ports **501** are directed to coupling system **801**. This coupling system comprises a polarisation conversion system **601** as described above. The beams of each of the input ports are focused at input circular focus point **803** by a single lens **802**. Anamorphic correction telescope **603** converts the circular beam at input circular focus point **803** to an elongated beam at input anamorphic focus point **804**. The angular encoding of the input ports in the focus at anamorphic focus point **804** is then passed through lens **805**. Anamorphic focus point **804** lies in the focal plane of lens **805**. Lens **805** is a cylindrical lens with optical power in the switching axis y. Lens **805** has no optical power in the dispersion axis, thus in FIG. **8b** the light signal passes through the lens **805** in the dispersion plane without deviation. Lens **805** thus acts to redirect the non-dispersed light signals incident on it into spatially separated parallel non-dispersed light signals. These parallel light signals are then passed through lenses **504** and **506** in which are in a 4F arrangement. Diffraction grating **505a** lies in the Fourier plane between the lenses **504** and **506**. The diffraction grating disperses the light signals incident on it into spatially separated spectral channels. The dispersed array of beams from the diffraction grating are incident on the first programmable deflection plane **508**.

[0115] The first programmable deflection plane **508** applies a programmable deflection to each spectral channel incident on it to form a deflected array of beams. The deflected array of beams is imaged through lenses **509** and **806** in a 4F arrangement. Diffraction grating **505b** lies in the Fourier plane between lenses **509** and **510**. Diffraction grating **505b** multiplexes the deflected spectra incident on it to form a multiplexed signal. That multiplexed signal is then passed through cylindrical lens **806** and imaged onto beam steering optical device **511**. Suitably, lenses **509** and **510** are the same as lenses **506** and **504**, diffraction grating **505b** is the same as diffraction grating **505a**, and lens **806** is the same as lens **805**.

[0116] The beam steering optical device **511** is implemented as two separate prisms **807a**, **807b**. Prism **807a** is positioned above the input anamorphic focus point **804**. Prism **807b** is positioned below the input anamorphic focus point **804**. The prisms **807a**, **807b** remap the beams incident on them to form a remapped array of beams. The prisms retroreflect the light incident on them and shift it to the side such that it is output above and below the output anamorphic focus point **606** for output to output ports **502**. Optionally, entrance micro-lenses **608a** are located in the optical path input to the prisms **807a**, **807b**. Optionally, exit micro-lenses **608b** are located in the optical path output from the prisms **807a**, **807b**. As in FIGS. **6a** and **6b**, these micro-lenses aid transmission if the depth of field of the lens **806** is not significantly greater than the required optical length.

[0117] The remapped array of beams output from the prisms **807a**, **807b** pass through telescope lenses **512** and **513**. These lenses image the remapped array of beams onto the second programmable deflection plane **514**. The second programmable deflection plane **514** applies a programmable deflection to the remapped array of beams, for example by use of a hologram on axis. The remapped array of beams deflected by the second programmable deflection plane **514** propagates through lens **515** which creates a single beam at output anamorphic focus point **606**. This elongated beam then passes through anamorphic telescope **605** to convert the elongated beam to a circular beam focused on output circular focus point **607**. This circular signal is then coupled to a selected output port **502** by coupling optics **516**.

[0118] In the implementation **900** of FIGS. **8a** and **8b** illustrated in FIG. **9**, the light signals travel from the input ports **501** through input anamorphic focus point **804** via lenses **805**, **504** and **506** and diffraction grating **505a** to the first programmable deflection plane **508**. The deflected array of beams deflected from the first programmable deflection plane **508** passes back through the same diffraction grating **505a** and lenses **506**, **504** and **805** to prisms **807**.

[0119] As described above, these prisms are positioned above and below the input anamorphic focus point **804** in the switching plane. FIG. **9** illustrates the dispersion plane. The prisms **807** direct the remapped array of beams to lens **512**. A mirror **901** (which is not shown in FIGS. **8a** and **8b**) is used in the implementation of FIG. **9** between the lens **512** and the lens **506**. This mirror **901** directs the remapped array of beams from the lens **512** to the second programmable deflection plane **514** via the lens **506**. In the implementation of FIG. **9**, the remapped array of beams is focused onto the second programmable deflection plane **514** via the lens **506** instead of the lens **513**. The array of beams deflected by the second programmable deflection plane **514** are then routed to the output ports **502** as described with respect to FIGS. **8a** and **8b**.

[0120] Although FIGS. **8a**, **8b** and **9** illustrate the prisms **807a**, **807b** as being positioned above and below the input anamorphic focus point **804**, in a further example the prisms **807a**, **807b** are instead positioned above and below the input circular focus point **803**. In this further example, the anamorphic telescope **605** is not used.

[0121] In either the example of FIGS. **8a**, **8b** and **9** or the further example mentioned above, a further anamorphic telescope may be positioned between the prisms **807a**, **807b** and the lens **512**. This further anamorphic telescope is used to enable different aspect ratios of the gaussian focus at the spectra **508** and the space switches **514**.

[0122] FIGS. **10a** and **10b** illustrate the detailed optical path of a further example of an add-drop WSS **1000**. FIG. **10a** illustrates the path in the switching plane, i.e. the y axis shown is the switching axis and the z axis is the optical axis. FIG. **10b** illustrates the optical path in the dispersion plane, i.e. the x axis shown is the dispersion axis and the z axis is the optical axis. Thus, FIG. **10b** illustrates an orthogonal plane to FIG. **10a**. The features of FIGS. **10a** and **10b** which are the same as those of previous figures and operate in the same manner are illustrated with the same reference numerals. The detail of those features is as described above. The structure and operation of the adWSS of FIGS. **10a** and **10b** differs from the description above by the following details.

[0123] The adWSS of FIGS. **10a** and **10b** does not use anamorphic telescopes. The light focused on the input circular focus point **803** is directed straight to the lens **805**. The light passes directly from

the lens **515** to lens **516**. The input circular focus point **803** forms the centre of the telecentric projection aligned with the two prisms **807a** and **807b**. The first prism **807a** is located above the input circular focus point **803**, and the second prism **807b** is located below the input circular focus point **803**.

[0124] In the example of FIGS. **10a** and **10b**, an asymmetric beam shape may be formed at the first and/or second programmable deflection planes. This is because the dispersion plane and the switching plane use different projections. The dispersion plane uses normal projection. The beam size in the dispersion plane is based on the width of the beam at the input circular focus point **803** in the dispersion direction. The beam size in the switching plane is determined by the divergence of the light beams at the input circular focus point **803** and the focal length of the cylindrical lens **805**. Thus, an anamorphic beam may be formed by causing a larger magnitude of projection in the switching plane *y* compared to the dispersion plane *x* at the first and/or second programmable deflection planes.

[0125] In the example of FIGS. **10a** and **10b**, the light path at the first and second programmable deflection planes is normal to the programmable deflection planes. Thus, any aberration effects and effects from the deflection, for example caused by the SLM hologram and pixel, are minimised across the spectrum. However, the on-axis undeflected light path is not viable, and hence an output port aligned with that undeflected path is not usable.

[0126] In the example described with respect to FIG. **5**, the array of space switches **518** is a vertical column that maps to the vertical column of spectra **517**. However, in an adWSS the number of drop ports may be significantly larger than the number of input ports. The space switches therefore take up much more vertical space on the programmable deflection plane than the spectra. Most SLM devices are rectangular in shape, and hence when the space switches and spectra are both in a column arrangement on the same SLM plane, the area of the SLM plane is not well utilised. This is shown in FIG. **4**, in which a large proportion of the area of the SLM plane is not used.

[0127] FIGS. **11**, **12** and **13** illustrate examples of adWSSs in which the arrangement of the input ports is different to that of the output ports. The beam steering optical device **511** remaps the deflected array of beams from the first programmable deflection plane **508** such that the combination of the images of the beams on the first programmable deflection plane **508** and the images of the remapped array of beams on the second programmable deflection plane **514** better utilises the area of the single SLM plane **507**. The beam steering optical device **511** may comprise a prism structure and/or a mirror structure that redirects the light incident on it in a controlled manner to achieve the remapping. The output ports are suitably distributed in the same way as the remapped array of beams on the second programmable deflection plane **514**.

[0128] In general, the angular switching by the first programmable deflection plane **508** is perpendicular to the spectral dispersion plane. In known systems, this confines the spectra and space switches to being in vertical columns. This limits the aspect ratio of the SLM panels which can be efficiently utilised, and thereby limits the performance of the system for a given SLM specification.

[0129] The following describes examples of the beam steering optical device **511** which can be used to remap a single column spectra array on an SLM panel to a multiple column space switch array on the same or different SLM panel, so as to better utilise the area of the SLM panel(s). This is done without requiring switching of the spectra channels in the dispersion plane. Instead, the beam steering optical device **511** comprises a prism structure and/or a mirror structure which has multiple sections, each of which provides a light path for a different portion of the single column spectra array. Those light paths are different, and result in the redistribution of the portions into different columns. The multiple columns are then imaged onto the space switches which have the corresponding multiple column configuration on the SLM panel.

[0130] The following also describes examples of the beam steering optical device **511** which can be used to remap multiple spectra columns on an SLM panel to a single vertical column space switch

array on the same or different SLM panel, so as to better utilise the area of the SLM panel(s). In the case of multiple spectra columns, switching can only be done perpendicular to the dispersion plane. In doing so with lenses only, not all space switches can be accessed by all the spectra channels. However, the beam steering optical device **511** comprises a prism structure and/or a mirror structure which has multiple sections, each of which provides a light path for the different columns of the multiple spectra columns. Those light paths are different, and result in the redistribution of the multiple columns into a single vertical column. The redistributed array is then passed through a lens to convert the redistributed array in the conjugate space switch plane, and imaged onto the space switches which have the corresponding single vertical column configuration on the SLM panel.

[0131] The features of FIGS. **11**, **12** and **13** which are the same as those of FIG. **5** and operate in the same manner are illustrated with the same reference numerals. The detail of those features is as described above with respect to FIGS. **5**, **6a** and **6b**. The structure and operation of the adWSSs of FIGS. **11**, **12** and **13** differs from the description of the adWSS of FIG. **5** by the following details.

[0132] In FIG. **11**, the input ports **1101** are arranged in a column array and the output ports **1102** are arranged in a grid array. The coupling lenses **503** match the distribution of the input ports **1101**. The lens array **516** matches the distribution of the output ports **1102**. The dispersed optical signals form spectra **517** on the first programmable deflection plane **508** in the same manner as described with respect to FIG. **5**. The deflected array of beams from the first programmable deflection plane **508** is in the same conformal shape as the arrangement of the input ports, i.e. a column array in this example. The deflected array of beams from the first programmable deflection plane **508** travels through a lens **509** and diffraction grating **505b** in the Fourier plane of lens **509**. The diffraction grating **505b** recombines the deflected spectra for each output port. An imaging lens **510** images the optical signal from the diffraction grating **505b** onto the beam steering optical device **511**. The beam steering optical device **511** remaps the deflected array of beams such that the remapped array of beams directed towards the second programmable deflection plane **514** matches the conformal shape of the output ports **1102**. The space switches **1103** on the second programmable deflection plane **514** match the conformal shape of the output ports. The arrangement of the space switches **1103** matches the conformal shape of the remapped array of beams. Thus, the optical device **511** remaps the vertical column of deflected beams input to it to a two-dimensional array of beams for imaging onto the space switches **1103** via the telescopic lenses **512** and **513**. The space switches **1103** then deflect the remapped array of beams to the output ports **1102** via lens **515** and coupling ports **516**.

[0133] The beam steering optical device may be implemented as a passive or active (i.e. electrically switchable) component. For example, a prism, mirror array, catadioptric system, freeform lens and mirror, a diffractive optical element, a diffractive optical element array, a holographic optical element, a holographic optical element array, a MEMS array or LCOS may be implemented. The MEMS array may be a DLP® (Digital Light Processing) MEMS array. The LCOS may be a Liquid Crystal based active beam steerer using switchable micro lenses.

[0134] Although the input ports **1101** are shown in a vertical column array, a different conformal arrangement of input ports may be used. For example, the input ports **1101** may be arranged as multiple columns. However, the spectra formed on the first programmable deflection plane must not overlap.

[0135] In FIG. **12**, the input ports **1201** are arranged in a grid array and the output ports **1202** are arranged in a column array. The coupling lenses **503** and **516** are as described above with respect to FIG. **11**. The input ports shown are in two columns. Thus, two columns of spectra **1204** are imaged onto the first programmable deflection plane **508**. These two columns of spectra are re-multiplexed through the 4F arrangement of lenses **509** and **510** separated by diffraction grating **505b**. The deflected re-multiplexed spectra array having two columns is remapped by the optical device **511** to a one-dimensional column array. This remapped single column array of beams is directed to the

second programmable deflection plane. In this example, the optical device **511** is a converter retroreflecting optical device. It allows access from all spectra on the first programmable deflection plane **508** to any space switch on the second programmable deflection plane **514** without requiring deflection in the dispersion axis of the spectra **1204**. In this example, lenses **512** and **513** are a conjugate converter instead of the imaging telescope of FIG. **11**. The space switches on the second programmable deflection plane **514** have the same column arrangement as the remapped array of beams. Thus the space switches deflect the remapped array of beams to the output ports **1202** via lens **515** and coupling ports **516**.

[0136] In FIG. **13**, the input ports **1301** are arranged in a grid array and the output ports **1302** are arranged in a different grid array. The coupling lenses **503** and **516** are as described above with respect to FIG. **11**. The input ports shown are in two columns. Thus, two columns of spectra **1304** are imaged onto the first programmable deflection plane **508**. These two columns of spectra are re-multiplexed through the 4F arrangement of lenses **509** and **510** separated by diffraction grating **505b**. The deflected re-multiplexed spectra array having two columns is remapped by the optical device **511** to an array having three columns to match the array of output ports. Optical device **511** comprises optical parts **1307** and **1309** and lens **1308**. First, optical part **1307** of optical device **511** converts the two column array of deflected beams from the first programmable deflection plane **508** to a column array. This column array is then passed through conjugate Fourier lens **1308** to second optical part **1309**. Second optical part **1309** converts the column array to a three column array matching the array of space switches on the second programmable deflection plane **514**. This three column array is then imaged by telescopic lenses **512** and **513** to the space switches **1305** on second programmable deflection plane **514**. Thus the space switches deflect the remapped array of beams to the output ports **1302** via lens **515** and coupling ports **516**.

[0137] The constituent optical parts **1307** and **1309** and lens **1308** may be separate components. Alternatively, **1307**, **1309** and **1308** may be a single component.

[0138] The examples of FIGS. **11**, **12** and **13** show that the arrangement of the individual space switches relative to the spectral planes can be remapped in order to better utilise the available space on the SLM panel. FIGS. **14a** and **14b** illustrate two examples in which the space switches have been redistributed in order that the spectra **1401** and space switches **1402** utilise a larger proportion of the area of the SLM panel **507** than in the prior art (see FIG. **4**). In both examples, both the spectra **1401** and space switches **1402** are in 2D arrays. This allows more space switches to be used for the same size of SLM panel compared to known devices. Additionally, this allows a greater height of spectra compared to known devices, thereby allowing more switchable positions. Thus, the capacity of the WSS increases for the same SLM panel area. In FIG. **14a**, the space switches and spectra are side by side. In FIG. **14b**, the space switches are below the spectra. Any combination of the space switches and spectra may be created using the remapping described herein. Suitably, the images on the first programmable deflection plane and the images of the remapped array of beams on the second programmable deflection plane cover over 80% of the area of the SLM panel. The images of the first programmable deflection plane and the remapped array of images on the second programmable deflection plane may cover the whole area of the SLM panel.

[0139] FIGS. **14a** and **14b** illustrate the SLM panel in a landscape orientation. This enables wide spectra which leads to better passband performance and greater total bandwidth than if narrower spectra were required due to an SLM panel being used in a portrait orientation.

[0140] However, an SLM panel in a portrait orientation may be utilised with any of the examples described herein.

[0141] FIGS. **15**, **16**, **17** and **18** illustrate some implementations of the beam steering optical device **511**. Each beam steering optical device remaps an array of points incident on it into a different conformal pattern. That different conformal pattern matches the geometrical distribution of the space switches on the second programmable deflection plane and the geometrical distribution of the image of the set of output ports on the second programmable deflection plane. The beam

steering optical device of FIG. 15 is a prism array 1500. The prism array 1500 remaps a single column array of beams to a multiple column array of beams, as shown in FIG. 11. Referring to FIGS. 11 and 15, the single column of spectra 517 are deflected by the first programmable deflection plane 508 and combined by the diffraction grating 505b. They are then imaged by lens 510 to form a set of column points 1501 at the entrance to the prism 1500. Microlenses 608a may optionally be used to collimate the light at the entrance of the prism 1500, as described above. The prism array 1500 remaps the column points 1501 to a grid of points 1503.

[0142] The prism array 1500 splits the column points 1501 into groups. FIG. 15 illustrates three groups 1501a, 1501b, 1501c, however it will be understood that more or fewer groups may be used. Group 1501c is underneath group 1501b which is underneath group 1501a in the single column of points 1501. Each group of points 1501a, 1501b, 1501c is directed to a different retroreflecting structure. Group 1501a is incident on structure 1502a, and exits structure 1502a as group 1503a. Group 1501b is incident on structure 1502b, and exits structure 1502b as group 1503b. Group 1501c is incident on structure 1502c, and exits structure 1502c as group 1503c. Prism structure 1502a causes group 1503a to be displaced to the side relative to group 1501a. Prism structure 1502b incorporates a periscope prism transmission which causes group 1503b to be displaced vertically relative to group 1501b. Prism structure 1502b also causes group 1503b to be displaced to the side relative to group 1501b. Thus, group 1503b is a different column of points to group 1503a which is positioned next to group 1503a. Prism structure 1502c causes group 1503c to be displaced to the side relative to groups 1503c. Prism structure 1502c incorporates a periscope prism transmission which causes group 1503c to be displaced vertically relative to group 1501c. Prism structure 1502c also causes group 1503c to be displaced to the side relative to group 1501c. Thus, group 1503c is a different column of points to groups 1503a and 1503b which is positioned next to group 1503b. In this way, the single column of points 1501 is remapped to a two-dimensional array of columns 1503.

[0143] Prism structures 1502a, 1502b and 1502c are non-overlapping and may be part of a single passive prism element.

[0144] If micro lenses 608a, 608b are not used, then the optical path through each prism structure for each group is altered such that the total optical path length for each group is the same. This ensures that focussing in the subsequent lens systems 512 and 513 would focus all the beams onto the second programmable deflection plane 514.

[0145] The prism array 1500 of FIG. 15 may be used as each of the prisms 807a, 807b described with respect to FIGS. 8a and 8b. An area between the prisms 807a, 807b is clear or transparent with no optical power so as to allow transmission of the input focus point 804 or 803.

[0146] The beam steering optical device of FIG. 16 is a mirror array 1600. The mirror array 1600 remaps a single column of points 1601 incident on it to a grid of points 1604. The mirror array 1600 splits the column points into groups. FIG. 16 illustrates three groups 1601a, 1601b, 1601c, however it will be understood that more or fewer groups may be used. Group 1601c is underneath group 1601b which is underneath group 1601a in the single column of points 1601. Each group of points 1601a, 1601b, 1601c follows a different optical path through the mirror array 1600. Each group of points reflects off a pair of mirror surfaces. Group 1601a reflects off mirror pair 1602a and 1603a as it passes through the mirror array 1600, exiting the mirror array 1600 as group 1604a. Group 1601b reflects off mirror pair 1602b and 1603b as it passes through the mirror array 1600, exiting the mirror array 1600 as group 1604b. Group 1601c reflects off mirror pair 1602c and 1603c as it passes through the mirror array 1600, exiting the mirror array 1600 as group 1604c. Each mirror surface of a pair of mirror surfaces are angled with respect to each other. The angle between the pair of mirror surfaces for each group of points is different. Thus, each group of points follows a different path through the mirror array 1600 thereby causing the geometrical distribution of the groups of points to change. Thus, mirror pair 1602a and 1603a causes group 1604a to be displaced to the side relative to group 1601a. Mirror pair 1602c and 1603c causes group 1604c to

be displaced vertically relative to group **1604c**, and also displaced to the side relative to group **1601c**. Thus, group **1604c** is a different column of points to group **1604a** which is positioned next to group **1604a**. Mirror pair **1602b** and **1603b** causes group **1604b** to be displaced to the side relative to group **1601b**, and also displaced vertically relative to group **1601b**. Thus, group **1604b** is a different column of points to groups **1604a** and **1604c** which is positioned next to group **1604c**. In this way, the single column of points **1601** is remapped to a two-dimensional array of columns **1604**.

[0147] A high depth of field, micro-lenses **608a**, **608b** or optical corrector plates may additionally be used to minimise the difference in optical paths between the groups **1604a**, **1604b**, **1604c**. This ensures that all the beams focus onto the second programmable deflection plane **514** after passing through subsequent lens systems **512** and **513**.

[0148] The examples of FIGS. **15** and **16** describe converting a column array into a grid array. It will be understood that inverting the described structures enables a grid array to be converted to a column array.

[0149] The beam steering optical device of FIG. **17** is a prism **1700**. The prism **1700** remaps a single column array **1701** by rotating it by 90° to form a row array **1702**. Prism **1700** comprises two orthogonal Amici Roof prisms **1703a** and **1703b**. If the beam steering optical device is to retroreflect the beam array, as in the examples of FIGS. **11** and **12**, then the prism **1700** includes a retroreflecting mirror **1704** at the output of the Amici Roof prisms. If the beam steering optical device is not to retroreflect the beam array, as in the example of FIG. **13**, then the prism **1700** does not include a retroreflecting mirror.

[0150] The beam steering optical device of FIG. **18** is an array of mirrored surfaces **1800**. The mirrored surface array **1800** remaps a single column array **1701** by rotating it by 90° to form a row array **1702**. Mirrored surface array **1800** comprises three differently angled reflecting surfaces **1801a**, **1801b** and **1801c**. These mirrored surfaces may be separate mirrors. Alternatively, the mirrored surfaces may be total internal reflection planes in a prism.

[0151] Although FIGS. **17** and **18** only illustrate a single column array **1701** which is rotated to form the row array **1702**, an array input to either device **1700** or device **1800** may comprise multiple columns. In this case, each column is rotated by 90° to form a row array. Thus, a two-dimensional array having X columns is remapped to a two-dimensional array having X rows.

[0152] The beam steering optical device implementations shown in FIGS. **17** and **18** can be used to redistribute space switches above or underneath the spectra on an SLM plane, as shown in FIG. **14b**.

[0153] In each of the beam steering optical devices described with respect to FIGS. **15**, **16**, **17** and **18**, the beam steering optical device remaps the deflected array of beams from the first programmable deflection plane by causing them to undergo two individual refractions, reflections or diffractions. The deflected array of beams has been transposed in two orthogonal directions. The deflected array of beams may have a linear distribution and the remapped array of beams a non-linear distribution. Alternatively, the deflected array of beams may have a non-linear distribution and the remapped array of beams a linear distribution. Alternatively, the deflected array of beams may have a non-linear distribution, and the remapped array of beams a different non-linear distribution. Alternatively, the deflected array of beams may have a linear distribution, and the remapped array of beams a different linear distribution, or a similar linear distribution but oriented in a different direction.

[0154] All of the examples described herein are directionless. Thus, they can operate in reverse. In other words, in each of the examples described above, light may enter the switch through the ports described as the output ports, and then travel through the internal structure of the switch in the reverse order to that described above, and leave the switch through the ports described as the input ports. For the adWSSs, this means light enters the switch via K add ports (the same as the output drop ports described above) and leaves via N output ports (the same as the input ports described

above). When a switch is operated in reverse to that described above, reciprocal control of the first and second programmable deflection planes is used. When operated in reverse, the light is first deflected by the second programmable deflection plane **514** described above, then routed via the beam steering optical device **511** to the first programmable deflection plane **508** before being output from the switch via the output ports.

[0155] Although the WSS examples described above all include spectra on a programmable deflection plane, the same beam steering optical device **511** may instead be used to remap an array of space switches on a first programmable deflection plane to a remapped array of space switches on a second programmable deflection plane. This enables a geometrical configuration of input ports to be mapped to a different geometrical configuration of output ports. All the channels in an input port are mapped to all the channels in an output port. In this case, no diffraction gratings are required in the switch.

[0156] The WSS devices described herein utilise a beam steering optical device **511** to remap the deflected array of beams from the first programmable deflection plane such that the remapped array of beams can be imaged (for example by magnification) so that it better utilises the area of the second programmable deflection plane. When the first and second programmable deflection planes are on the same SLM panel, this means that the distribution of spaces switches and/or spectra of each programmable deflection plane efficiently covers the area of the SLM panel. This enables multiple WSSs to use the same SLM panel. When the first and second programmable deflection planes are on separate SLM panels, the remapping performed by the beam steering optical device **511** enables a smaller or cheaper second SLM to be used for the second programmable deflection plane.

[0157] SLM panels are generally rectangular. Thus, the remapping described herein usefully changes the conformal structure of the deflected beams from the first programmable deflection plane from a column distribution to another column distribution or a grid distribution. However, the beam steering optical devices described herein independently control the spatial positioning and/or orientation of each beam from the deflected array of beams in the remapped array of beams. Thus, each beam is remapped independently of every other beam in the deflected array. This remapping is in two dimensions. This enables the overall distribution of the deflected array to be remapped to any distribution. That remapped distribution may be linear or non-linear. It may be columnar in shape. It may have a non-column shape. For example, the remapped distribution may be hexagonal. Thus, any shaped SLM panel, be it rectangular, non-rectangular or irregular shaped can be better filled utilising the beam steering optical devices described herein. The ability to remap to any configuration allows flexibility in the LCOS aspect ratio to meet a particular design target. Switching in one or two dimensions can be implemented.

[0158] It is also noted that the deflected array of beams input to the beam steering optical devices described herein need not be in a column arrangement. A non-column distribution of beams can be input to the beam steering optical device and it remap that non-column distribution of beams into whatever distribution of beams is needed for the second programmable deflection plane, be that a column arrangement or not.

[0159] The remapping performed by the beam steering optical device may be controlled so as to generate a distribution of spectra/space switches on the second programmable deflection plane which improves the coupling of light to the output ports, for example by reducing crosstalk.

[0160] The examples described herein utilise one-dimensional angular deviation at the first programmable deflection plane in a direction normal to the dispersion plane. Angle switching in the dispersion plane is not required. However, angular deviation in the dispersion plane may also be implemented at the first programmable deflection plane. A small dispersion axis shift on the spectra creates multiple columns. This may be used if a non-column arrangement is desired for the remapped array of beams output from the beam steering optical device. By incorporating a small dispersion axis shift, the maximum angle required for a given number of ports reduces.

[0161] The beam steering optical devices described herein remap the deflected array of beams from the first programmable deflection plane. For an adWSS, the image of the re-multiplexed spectra and space switches are in conjugate planes since the space switches are chosen by altering the angle in the spectral plane. Thus, the conjugate plane must be changed on reflection. At least one imaging element between the spectral plane and the space switch plane is in the conjugate configuration. This imaging element may be a lens or mirror. This imaging element may form part of the beam steering optical device or be separate from it. The same applies for a M×N switch.

[0162] It is preferable to minimise the number of distinct components in the switch and for the construction and layout of those components to be as simplified as possible. This enables the switch to be more compact and more straightforward to manufacture.

[0163] The examples described above utilise a “fan-in” approach for imaging the input ports onto the first programmable deflection plane. A fan-in approach is one in which the optical signals from the input ports are brought to a common point focus before reaching the first programmable deflection plane. In the examples described above this point focus is the input circular focus point or the input anamorphic focus point.

[0164] The examples described below additionally utilise a “fan-out” approach, in which the optical signals from the second programmable deflection plane are brought to a common point focus before reaching the output ports. It is known to use a fan-in or a fan-out approach, but not both, for the reason explained below. A fan-in approach, as described in the examples above, is preferred to a fan-out approach because this uses fewer components overall in the switch. However, utilising both a fan-in and fan-out approach enables optimum operation for the single SLM device which comprises the first and second programmable deflection planes. It also enables fewer, simpler components to be used in the switch which leads to a more compact switch which is easier to manufacture.

[0165] In the optical path between the first and second programmable deflection planes there is an image plane of the first programmable deflection plane and an image plane of the second programmable deflection plane which are in conjugate planes. There is spatial data in both of these conjugate planes. Because of this, in the examples described above, there is no simple plane between the first and second programmable deflection planes in which the optical signals come to a single focus point.

[0166] The examples below describe modifications to the internal components of the switches described above to enable both a fan-in and fan-out approach to be used. In these examples, a gap is created in the conjugate image plane of the first programmable deflection plane between a first group of beams projected on that conjugate image plane by the first programmable deflection plane and a second group of beams projected on that conjugate image plane by the first programmable deflection plane. The second programmable deflection plane deviates all the optical signals incident on it towards the output ports through the gap in this conjugate plane. Thus, the switch implements a fan-out procedure as well as a fan-in procedure.

[0167] FIGS. **19a** and **19b** illustrate the detailed optical path of an exemplary add-drop WSS **2300**. FIG. **19a** illustrates the path in the switching plane, i.e. the y axis shown is the switching axis and the z axis is the optical axis. FIG. **19b** illustrates the optical path in the dispersion plane, i.e. the x axis shown is the dispersion axis and the z axis is the optical axis. Thus, FIG. **19b** illustrates an orthogonal plane to FIG. **19a**. The features of FIGS. **19a** and **19b** which are the same as those of FIGS. **8a** and **8b** and operate in the same manner are illustrated with the same reference numerals. The detail of those features is as described above with respect to FIGS. **8a** and **8b**. The structure and operation of the adWSS of FIGS. **19a** and **19b** differs from the description of the adWSS of FIGS. **8a** and **8b** above by the following details.

[0168] The input ports **2301** of FIGS. **19a** and **19b** are different to those of FIGS. **8a** and **8b**. The input ports **2301** are distributed such that there is a gap in the beams incident on the first programmable deflection plane **508** from the input ports **2301**. This gap may be generated by

removing one or more ports from the set of input ports **501** of the example of FIGS. **8a** and **8b**. Alternatively, this gap may be generated by altering the arrangement of input ports **501** of the example of FIGS. **8a** and **8b**. The set of input ports **2301** thus comprises a first set of input ports **2301a** and a second set of input ports **2301b**. The first set of input ports **2301a** is spatially separated from the second set of input ports **2301b** by a spacing **2302**, so as to form a gap between those beams incident on the first programmable deflection plane **508** from the first set of input ports **2301a** and those beams incident on the first programmable deflection plane **508** from the second set of input ports **2301b**.

[0169] The light signals propagate from the input ports **2301** to the first programmable deflection plane **508** as described with reference to FIGS. **8a** and **8b**. The light incident on the first programmable deflection plane is normal to the first programmable deflection plane. This minimises insertion loss and utilises the best operation of the first programmable deflection plane, particularly in the case that the first programmable deflection plane is an LCOS device.

[0170] FIG. **20** illustrates the arrangement of spectra and space switches in the case that the switch is an adWSS and both the first and second programmable deflection planes are implemented on a single SLM device. The spectra **2401** are arranged in a first group **2401a** and a second group **2401b**. The first and second groups of spectra are separated by gap **2302**. This gap is shown as centrally located on the first programmable deflection plane. However, it could be located elsewhere on the first programmable deflection plane. In this example, the gap **2302** is the same size as a spectrum. It is not formed by splitting any of the spectra. The spectral distribution on the first programmable deflection plane **508** of FIGS. **19a** and **19b** is therefore different to that shown in FIGS. **14a** and **14b**. Although shown as a landscape arrangement, it will be understood that the SLM device may be arranged in a portrait arrangement or have a non-rectangular shape, as described above.

[0171] In FIG. **20**, the gap **2302** is an inactive area on the single SLM. Alternatively, the gap **2302** may be used to accommodate space switches. Suitably, these space switches have been remapped to be located in the gap by the beam steering optical device **511** described above. Alternatively, if the first programmable deflection plane is implemented on a first SLM device, and the second programmable deflection plane implemented on a second SLM device, then the gap **2302** may be a gap between the first and second SLM devices.

[0172] The first programmable deflection plane **508** applies a programmable deflection to each spectral channel incident on it to form a deflected array of beams. The deflected array of beams comprises a first set of deflected beams separated from a second set of deflected beams by a gap. The deflected array of beams is imaged back through the same lenses **506** and **504** in the 4F arrangement. The deflected array of beams passes through the same diffraction grating **505a**. Diffraction grating **505a** multiplexes the deflected spectra incident on it to form a multiplexed signal. The multiplexed signal passes through the same cylindrical lens **805**. The multiplexed signal is focussed to a point above the input anamorphic focus point **804**. The multiplexed signal then passes through anamorphic correction telescope **603** to beam steering optical device **511**.

[0173] Beam steering optical device **511** is optional in this WSS, but if incorporated operates as described in any of the examples above.

[0174] The beams output from the beam steering optical device **511** are still in two groups separated by a gap. These beams then pass through lenses **2304** and **506** in a telescope arrangement for focussing on the second programmable deflection plane **514**. The optical signal incident on the second programmable deflection plane is normal to the second programmable deflection plane.

[0175] Referring to FIG. **20**, the space switches **2402** are arranged in a first group **2402a** and a second group **2402b**. The first and second groups of space switches are separated by gap **2403**. This gap is shown as centrally located on the second programmable deflection plane. However, it could be located elsewhere on the second programmable deflection plane. This gap **2403** is not required to enable the switch to implement a fan-out arrangement. However, implementing a gap on both the

first and second programmable deflection planes simplifies the optics in the remainder of the switch.

[0176] Referring back to FIG. **19a**, the second programmable deflection plane **514** applies a programmable deflection to each of the first and second sets of beams incident on it. The deflection applied by the second programmable deflection plane **514** is so as to direct all the beams incident on it to a gap in the beams projected on a conjugate image plane of the first programmable deflection plane. The beams projected on this image plane from the first programmable deflection plane include a first set of deflected beams and a second set of deflected beams separated by the gap. The deflected first and second sets of beams from the second programmable deflection plane **514** pass through lens **506** and onto mirror **2305** located in the gap.

[0177] Mirror **2305** is located in the optical path between the first and second programmable deflection planes. Mirror **2305** is located in the optical path between the second programmable deflection plane and the output ports. Mirror **2305** is located where the gap is between the first and second sets of deflected beams. Mirror **2305** is located in the focal plane of lens **506**. Mirror **2305** is also located in the focal plane of lens **2304**. The mirror **2305** is angled so as to deflect the deflected first and second sets of beams from the second programmable deflection plane **514** towards the set of output ports. The mirror **2305** directs the deflected beams towards telescope **2306**. Telescope **2306** may be implemented as two lenses. Telescope **2306** adjusts the magnification to a focus plane **2307**. The magnified deflected beams are then coupled to a selected output port **502** by coupling optics **516**. The output port **502** selected is determined by the angle of deflection from the second programmable deflection plane as described above.

[0178] The lens **506** is in the optical path four times in this WSS: between the input ports and the first programmable deflection plane, between the first programmable deflection plane and the beam steering optical device, between the beam steering optical device and the second programmable deflection plane, and between the second programmable deflection plane and the output ports.

[0179] Although structure **2305** has been described herein as a mirror, it may alternatively be any other suitable beam steering device. For example, structure **2305** may be any of a mirror, mirror array, prism total internal reflection surface, diffractive optical element and holographic optical element.

[0180] The implementation described with reference to FIGS. **19a** and **19b** is simple to implement. However, the SLM area is not as efficiently used by virtue of the unused area of the gap **2302**.

[0181] FIGS. **21a** and **21b** illustrate the detailed optical path of a further exemplary add-drop WSS **2500** which enables both a fan-in and fan-out approach. FIG. **21a** illustrates the path in the switching plane, i.e. the y axis shown is the switching axis and the z axis is the optical axis. FIG. **21b** illustrates the optical path in the dispersion plane, i.e. the x axis shown is the dispersion axis and the z axis is the optical axis. Thus, FIG. **21b** illustrates an orthogonal plane to FIG. **21a**. The features of FIGS. **21a** and **21b** which are the same as those of FIGS. **19a** and **19b** and operate in the same manner are illustrated with the same reference numerals. The detail of those features is as described above with respect to FIGS. **19a** and **19b**. The structure and operation of the adWSS of FIGS. **21a** and **21b** differs from the description of the adWSS of FIGS. **19a** and **19b** above by the following details.

[0182] Firstly, the input ports **2501** of FIGS. **21a** and **21b** are the same as those of FIGS. **8a** and **8b**. They are not distributed so as to create a gap in the beams incident on the first programmable deflection plane **508** as with FIGS. **19a** and **19b**. The light signals propagate from the input ports **2501** to the first programmable deflection plane **508** as described with reference to FIGS. **8a** and **8b**. The light incident on the first programmable deflection plane is normal to the first programmable deflection plane.

[0183] FIG. **22** illustrates the arrangement of spectra and space switches in the case that the switch is an adWSS and both the first and second programmable deflection planes are implemented on a single SLM device. The spectra **2601** are arranged so as to utilise the whole height of the SLM

device. There is no unused area of the first programmable deflection plane. Although shown as a landscape arrangement, it will be understood that the SLM device may be arranged in a portrait arrangement or have a non-rectangular shape, as described above.

[0184] The first programmable deflection plane **508** applies a programmable deflection to each spectral channel incident on it to form a deflected array of beams. This deflected array of beams propagates through to beam steering optical device **511** as described with respect to FIGS. **19a** and **19b**.

[0185] Again, beam steering optical device **511** is optional in this WSS, but if incorporated operates as described in any of the examples above.

[0186] The beams output from the beam steering optical device **511** pass through lens **2304** to optical structure **2502**. Optical structure **2502** is located in the optical path between the first and second programmable deflection planes. Optical structure **2502** is located in the optical path between the second programmable deflection plane and the output ports. Optical structure **2502** is located in the focal plane of lens **506**. Optical structure **2502** is also located in the focal plane of lens **2304**. Optical structure **2502** is located in the conjugate image plane of the first programmable deflection plane. Optical structure **2502** alters the configuration of beams incident on it from the first programmable deflection plane so as to generate a gap between a first set of those beams and a second set of those beams. The optical structure **2502** may comprise a prism or prism assembly or a mirror or mirror assembly. Exemplary optical structures **2502** will be described in more detail with reference to FIGS. **25** and **26** below.

[0187] The first and second sets of beams output from the optical structure **2502** are incident on lens **506**. Lenses **2304** and **506** are in a 4F arrangement, with optical structure **2502** in the Fourier plane between lenses **2304** and **506**. The optical signal incident on the second programmable deflection plane is normal to the second programmable deflection plane.

[0188] Referring to FIG. **22**, the space switches **2602** are arranged so as to utilise the whole height of the SLM device. There is no gap in the space switches.

[0189] The second programmable deflection plane **514** applies a programmable deflection to each of the first and second sets of beams incident on it. The deflection applied by the second programmable deflection plane **514** is so as to direct all the beams incident on it through the gap created by the optical structure **2502**. Thus, the deflected first and second sets of beams from the second programmable deflection plane **514** pass through lens **506** and through optical structure **2502** via the gap. From there, the deflected beams are directed towards telescope **2306**, and through coupling optics **516** to output ports **502** as described with reference to FIGS. **19a** and **19b**.

[0190] FIG. **23** illustrates an implementation **2700** of the arrangement of FIGS. **21a** and **21b** in the dispersion plane. The detail of the features of FIG. **23** is as described above with respect to FIGS. **21a** and **21b**. In the implementation of FIG. **23**, a single device comprises both the first and second programmable deflection planes. This single device may, for example, be a SLM device.

[0191] Optical signals travel from the input ports **2501** through polarisation conversion system **601** to input circular focus point **803**. An anamorphic converter converts the circular beam to an elongated beam which is focussed at input anamorphic focus point. The optical signals then passes through lens **805** and then through lenses **504** and **506** which are in a 4F arrangement. Diffraction grating **505a** in the Fourier plane between lenses **504** and **506** disperses the optical signals into spatially separated spectral channels, which are incident on the first programmable deflection plane **508**.

[0192] The deflected array of beams deflected from the first programmable deflection plane **508** passes back through the same diffraction grating **505a**, which this time multiplexes the beams, and lenses **506**, **504** and **805** to beam steering optical device **511**. Beam steering optical device is positioned above and/or below the input circular focus point **803** in the switching plane. Beam steering optical device **511** directs the remapped array of beams to lens **2304**. Lens **2304** converts the remapped array of beams received from the beam steering optical device to parallel light

signals, which it directs to the optical structure **2502** in the Fourier plane of lens **2304**. A mirror **2701** (which is not shown in FIGS. **21a** and **21b**) is used in the implementation of FIG. **23** between lens **2304** and optical structure **2502**. This mirror **2701** directs the remapped array of beams from the lens **2304** to the optical structure **2502**. Suitably, this mirror **2701** is a flat mirror.

[0193] The optical structure **2502** generates a gap in the remapped array of beams as described above, and directs the remapped array of beams to the second programmable deflection plane **514** on the single device **507** via the lens **506**. The array of beams deflected by the second programmable deflection plane **514** is then routed through the gap in the optical structure **2502** and onto the output ports **502** via the telescope **2306** and coupling optics **516** as described above.

[0194] Although FIG. **23** illustrates the beam steering optical device as being positioned above and/or below the input circular focus point **803** in the switching plane, the beam steering optical device could instead be positioned above and/or below the input anamorphic focus point **804** in the switching plane. In this case, an anamorphic telescope is included in the optical path between the second programmable deflection plane and the output ports. For example, the anamorphic telescope may be included between the telescope **2306** and the coupling ports **516**. In either the example of FIGS. **21a**, **21b** and **23** or this further example, a further anamorphic telescope may be positioned between the beam steering optical device **511** and the lens **2304**. This further anamorphic telescope is used to enable different aspect ratios of the gaussian focus at the spectra on the first programmable deflection plane **508** and the space switches on the second programmable deflection plane **514**.

[0195] FIG. **24** illustrates an implementation **2800** of the arrangement of FIGS. **21a** and **21b** in the dispersion plane. The detail of the features of FIG. **24** is as described above with respect to FIGS. **21a** and **21b**. In the implementation of FIG. **24**, a single device comprises both the first and second programmable deflection planes. This single device may, for example, be a SLM device. The implementation **2800** of FIG. **24** utilises two mirrors **2801** and **2802**. These mirrors are curved mirrors. Mirror **2801** performs the function of lenses **504** and **506** of FIGS. **21a** and **21b**. Lens **2802** performs the function of lens **2304** of FIGS. **21a** and **21b**.

[0196] Optical signals travel from the input ports **2501**, through anamorphic converter **603** to lens **805**. The optical signals are then incident on first mirror **2801**, which reflects them to diffraction grating **505a**. Diffraction grating **505a** is positioned one focal length of the first mirror **2801** from the first mirror **2801**. Diffraction grating **505a** disperses the optical signals into spatially separated channels, which are then directed back to the first mirror **2801**. The first mirror **2801** reflects the dispersed optical signals to the first programmable deflection plane **508** on device **507**. The first programmable deflection plane **508** is positioned one focal length of the first mirror **2801** from the first mirror **2801**.

[0197] The deflected array of beams deflected from the first programmable deflection plane **508** are directed back to the first mirror **2801**. The first mirror **2801** reflects the deflected array of beams back to the diffraction grating **505a**, which this time multiplexes the beams. From the diffraction grating **505a**, the multiplexed beams are directed to the first mirror **2801**. The first mirror **2801** reflects the multiplexed beams to the beam steering optical device **511**. Beam steering optical device **511** remaps the multiplexed beams to form a remapped array, which it directs to a second mirror **2802**.

[0198] The second mirror **2802** is positioned one of its focal lengths from the beam steering optical device **511**. The second mirror **2802** is positioned one of its focal lengths from the optical structure **2502**. The second mirror **2802** reflects the remapped array of beams from the beam steering optical device **511** to the optical structure **2502**. The optical structure **2502** generates a gap in the remapped array of beams as described above, and directs the remapped array of beams to the first mirror **2801**. The optical structure **2502** is positioned one focal length of the first mirror **2801** from the first mirror **2801**. The first mirror **2801** reflects the remapped array of beams to the second programmable deflection plane **514** on the single device **507**.

[0199] The array of beams deflected by the second programmable deflection plane **514** is directed to the first mirror **2801**. The first mirror **2801** reflects the array of beams deflected by the second programmable deflection plane **514** to the gap in the optical structure **2502** and onto the output ports **502** via the telescope **2306** and coupling optics **516** as described above.

[0200] The example of FIG. **24** has a minimal number of separate components to perform the described functions of the adWSS.

[0201] FIGS. **25** and **26** each illustrates an implementation of the optical structure **2502**. In both of these implementations, the optical structure **2505** alters the configuration of the beams incident on it so as to generate a gap between those beams.

[0202] The optical structure **2502** of FIG. **25** comprises a prism or mirror array **2900**. The prism/mirror array **2900** comprises a first mirror assembly **2903a,2903b** and a second mirror assembly **2904a, 2904b**. Optical signals **2901** entering the prism/mirror array **2900** are incident on the first mirror assembly **2903a, 2903b**. First mirror assembly comprises a first mirror **2903a** and a second mirror **2903b**. A portion of the optical signals **2901** entering the prism/mirror array **2900** are incident on the first mirror **2903a**. A portion of the optical signal **2901** entering the prism/mirror array **2900** are incident on the second mirror **2903b**. The first and second mirrors **2903a,2903b** are angled away from each other. Thus, when parallel optical signals are incident on the first and second mirrors, the first group of optical signals reflected by the first mirror diverge from the second group of optical signals reflected by the second mirror. A gap thereby forms between the divergent first and second groups of optical signals. These non-parallel divergent first and second groups of optical signals are then routed internally in the prism to the second mirror assembly **2904a,2904b**.

[0203] Second mirror assembly comprises a third mirror **2904a** and a fourth mirror **2904b**. The second mirror assembly realigns the first and second groups of divergent optical signals output from the first mirror assembly to be parallel to each other. Third mirror **2904a** receives the first group of optical signals reflected from the first mirror **2903a**, and reflects this first group of optical signals out of the prism/mirror array. Fourth mirror **2904b** receives the second group of optical signals reflected from the second mirror **2903b**, and reflects this second group of optical signals out of the prism/mirror array in a direction parallel to the first group of optical signals. The third mirror **2904a** and fourth mirror **2904b** are inverse to each other. The first and second group of optical signals output from the prism/mirror array **2902** are parallel and separated from each other by a gap. The propagation of light signals **2905** from the second programmable deflection plane to the output ports passes through this gap.

[0204] Although the light incident on the prism/mirror array **2901** is shown as parallel to and in the same direction as the light exiting the prism/mirror array **2902**, the mirrors of the first and second mirror assemblies may be arranged to direct the light in any direction. For example, the light **2902** may exit the prism/mirror array in an opposing direction to the direction it is input in.

[0205] The first and second mirror assemblies are non-overlapping and may be part of a single passive prism element.

[0206] FIG. **26** illustrates an optical structure which is a prism **3000** having total internal reflection surfaces which perform the same function as the mirrors of FIG. **25**. Thus, parallel incident light beams **3001** on the prism **3000** are redirected within the prism such that the light beams **3002** exiting the prism are in two groups which are parallel but separated by a gap.

[0207] The prism comprises a first part having a first total internal reflection (TIR) surface **3003a** and a second total internal reflection surface **3004a**. The prism also comprises a second part having a third total internal reflection surface **3003b** and a fourth total internal reflection surface **3004b**. A portion of the optical signals **3001** entering the prism is incident on the first TIR surface **3003a**. A portion of the optical signals **3001** entering the prism is incident on the third TIR surface **3003b**. The first and third TIR surfaces are angled away from each other. Thus, when parallel optical signals are incident on the first and third TIR surfaces, the first group of optical signals reflected by

the first TIR surface **3003a** diverge from the second group of optical signals reflected by the third TIR surface **3003b**. A gap thereby forms between the divergent first and second groups of optical signals. These non-parallel divergent first and second groups of optical signals are then routed internally in the prism.

[0208] The first group of signals reflected by the first TIR surface **3003a** is incident on the second TIR surface **3004a**. The second TIR surface **3004a** reflects this first group of signals out of the prism. The second group of signals reflected by the third TIR surface **3003b** is incident on the fourth TIR surface **3004b**. The fourth TIR surface **3004b** reflects this second group of signals out of the prism. The second and fourth TIR surfaces are angled with respect to each other to realign the first and second groups of divergent optical signals from the first and third TIR surfaces such that they are parallel to each other when output from the prism **3000**. The gap between the first and second groups of optical signals caused by their diverging after being reflected by the first and third TIR surfaces is retained when they reflect from the second and fourth TIR surfaces. Thus, the first and second groups of optical signals output from the prism **3002** are separated by a gap. The propagation of light signals **3005** from the second programmable deflection plane to the output ports passes through this gap.

[0209] Although the light incident on the prism **3001** is shown as parallel to and in the same direction as the light exiting the prism **3002**, the TIR surfaces may be angled and positioned with respect to each other to direct the light in any direction. For example, the light **3002** may exit the prism in an opposing direction to the direction it is input in.

[0210] The prism **3000** illustrated in FIG. **26** lies in the image plane of the first programmable deflection plane but wholly external to the gap in the beams projected on that image plane from the first programmable deflection plane. The light signals from the second programmable deflection plane which pass through the gap in the image plane are not incident on the prism **3000** and hence do not propagate through it. Alternatively, the prism **3000** may have a section which is located at the gap in the beams projected on the image plane from the first programmable deflection plane. In this case, that section has no optical power. The section therefore does not obstruct or deflect the light signals from the second programmable deflection plane as they pass through it.

[0211] The examples described with reference to FIGS. **19a** to **26** enable the beams deflected from the second programmable deflection plane to be directed to the output ports through a gap in the beams projected from the first programmable deflection plane on a conjugate image plane. This allows those beams deflected from the second programmable deflection plane to pass through a single point as they are output, i.e. a fan-out approach. This enables fewer components to be used in the switch overall, and the use of a simplified mirror arrangement such as that shown in FIG. **24**. The switch is therefore more compact and simpler and cheaper to manufacture. It also benefits from reduced insertion loss whilst maintaining CDC behaviour.

[0212] In the examples described with reference to FIGS. **19a** to **26**, entrance micro-lenses **608a** located in the optical path input to the beam steering optical device, and/or exit micro-lenses **608b** located in the optical path output from the beam steering optical device may optionally be used. As in FIGS. **6a** and **6b**, these micro-lenses aid transmission.

[0213] In the examples described with reference to FIGS. **19a** to **26**, a beam steering optical device **511** is shown in addition to the optical arrangement which enables the fan-out approach to be used. However, the beam steering optical device **511** is not essential to these switches. It is an optional device used to enable the remapping described herein. The beam steering optical device **511** may be omitted from any of the WSSs shown in FIGS. **19a** to **26**, in which case that WSS has a fan-in and fan-out set up without the remapping described herein.

[0214] FIG. **27** illustrates an add-drop WSS **3100** in the dispersion plane which incorporates both the remapping described herein and the creation of a gap in the image plane of the first programmable deflection plane described herein. The features of FIG. **27** which are the same as those described above and operate in the same manner are illustrated with the same reference

numerals.

[0215] Optical signals travel into the WSS via input ports **3101**. Input ports **3101** may comprise a non-linear distribution in the dispersion plane. For example, there may be a plurality of columns of ports in the dispersion plane. Alternatively, there may be another non-linear distribution in the dispersion plane such as a hexagonal distribution.

[0216] The optical signals travel from the input ports **3101** through coupling optics **801**. Coupling optics **801** comprises coupling lenses and polarisation converters, which function as described previously herein. The optical signals travel from the coupling optics **801** to anamorphic telescope **603**. Anamorphic telescope **603** causes the optical signals to form a focus at input anamorphic focus point **804**. The optical signals are modified by the anamorphic telescope **603** so as to have different widths in the dispersion axis x and switching axis y as described herein.

[0217] The optical signals pass from the anamorphic telescope **603** through cylindrical lens **805**. Cylindrical lens **805** has power in the switching axis y but no power in the dispersion axis x. Cylindrical lens **805** has a focal length equal to its distance from the input anamorphic focus point **804**. The optical signals are collimated by lens **805** and passed to mirror **3102**.

[0218] The mirror **3102** is curved. The mirror **3102** may be spherically or cylindrically curved (with power in the dispersion axis). Mirror **3102** has a focal length equal to its distance from the input anamorphic focus point **804**. The optical signals reflect from mirror **3102** towards diffraction grating **505a**. Diffraction grating **505a** disperses the optical signals into spatially separated spectral channels, which are then directed back to mirror **3102**. The dispersed optical signals are reflected by mirror **3102** towards SLM **507** to produce spectra on the first programmable deflection plane **508** of the SLM **507**.

[0219] The first programmable deflection plane **508** deflects the dispersed optical signals incident on it in the switching plane. Thus, the first programmable deflection plane **508** adds an angular deviation to the dispersed optical signals in the switching plane. The first programmable deflection plane **508** directs the deflected dispersed optical signals to the mirror **3102**. The mirror **3102** directs the deflected dispersed optical signals to the diffraction grating **505a** where they are multiplexed by the diffraction grating **505a** and directed back to the mirror **3102**. The mirror **3102** directs the multiplexed optical signals through lens **805** to flat mirror **3103**. Flat mirror **3103** is located in the same position as the input anamorphic focus point **804** in the dispersion plane shown in FIG. 27, but above or below the input anamorphic focus point **804** in the switching plane y. The deviation applied by the first programmable deflection plane **508** shifts the focus position of the optical signals onto the flat mirror **3103**.

[0220] The flat mirror **3103** reflects the multiplexed optical signals to mirror **3102** through cylindrical lens **805**. Mirror **3102** then reflects the multiplexed optical signals to a first composite mirror **3104**. First composite mirror **3104** is shown in more detail in FIG. 28a. It comprises two flat mirrors **3105a** and **3105b** and anamorphic telescope **3106**. The flat mirrors **3105a** and **3105b** may be in a retroreflecting arrangement with the anamorphic telescope **3106** in the optical path between them. The input and output angles of each optical signal to the flat mirrors **3105a** and **3105b** may differ. The anamorphic telescope **3106** changes the magnification in the dispersion axis x. The anamorphic telescope **3106** controls the relative magnification of the first programmable deflection plane **508** (spectra) and the second programmable deflection plane **514** (space switches), and therefore the overall profile height.

[0221] An alternative first composite mirror **3104** is shown in more detail in FIG. 28c. It comprises two curved mirrors **3114a** and **3114b** with power in the dispersion direction, thereby forming part of an anamorphic telescope. Two further lenses **3115a** and **3115b** of the anamorphic telescope are positioned relative to the mirrors **3114a** and **3114b** to achieve the same overall effect as the arrangement of FIG. 28a. Lens **3115a** is in the optical path between curved mirrors **3114a** and **3114b**. Lens **3115b** is in the optical path of beams reflected from mirror **3114b**.

[0222] From the first composite mirror **3104**, the optical signals are directed to the mirror **3102**.

The mirror **3102** then directs the optical signals to a first optical part **1307** of a beam steering optical device. The first optical part **1307** remaps the beams incident on it which are arranged as a plurality of columns to a single column arrangement, as described above with respect to FIG. **13**. The first optical part **1307** outputs the remapped single column arrangement of beams to the mirror **3102**.

[0223] The mirror **3102** directs the remapped single column arrangement of beams to the first composite mirror **3104**. The first composite mirror **3104** reflects the remapped single column arrangement of beams back to the mirror **3102**. The mirror **3102** then directs the single column arrangement of beams to a second optical part **1309** of the beam steering optical device. The second optical part **1309** remaps the single column arrangement of beams into a remapped distribution. This remapped distribution matches the output port distribution. The second optical part **1309** outputs the remapped optical signals back to mirror **3102**.

[0224] The mirror **3102** then directs the remapped optical signals to the first composite mirror **3104**, which directs the remapped optical signals back to the mirror **3102**. The mirror **3102** then directs the remapped optical signals to the optical structure **2502**. The optical structure **2502** operates as described above to form a gap in the beams incident on it, and hence a gap in the spectral plane at the location of the optical structure **2502**. The optical signals output from the optical structure **2502** propagate to a third mirror **3110**. Third mirror **3110** is different to the mirror **3102** and oriented differently to the mirror **3102**. Suitably, the third mirror **3110** has the same optical power as the mirror **3102**. Third mirror **3110** focusses the optical signals onto a second composite mirror **3111**.

[0225] The second composite mirror **3111** may be in the plane of the diffraction grating **505a**. The second composite mirror **3111** may be adjacent to the diffraction grating **505a**. The second composite mirror **3111** may be attached to the diffraction grating **505a**. The second composite mirror **3111** is shown in more detail in FIG. **28b**. It comprises two flat mirrors **3112a** and **3112b** and anamorphic telescope **3113**. The flat mirrors **3112a** and **3112b** may be in a retroreflecting arrangement with the anamorphic telescope **3113** in the optical path between them. The input and output angles of the optical signals to the flat mirrors **3112a** and **3112b** may differ. The anamorphic telescope **3113** changes the magnification in the dispersion axis x . The anamorphic telescope **3113** corrects the distribution to fit on the second programmable deflection plane **514**. Alternatively, the second composite mirror **3111** may take the form shown in and described above with respect to FIG. **28c**.

[0226] From the second composite mirror **3111**, the optical signals are directed to the mirror **3102** such that the mirror **3102** reflects the optical signals to form space switches on the second programmable deflection plane **514**. The second programmable deflection plane **514** deflects the optical signals back to the mirror **3102**. The mirror **3102** then directs the optical signals to the second composite mirror **3111**. The second composite mirror **3111** then directs the optical signals to the third mirror **3110**. The third mirror **3110** directs the optical signals to the optical structure **2502**. The optical signals pass through the gap in the image plane where the optical structure **2502** is positioned to the anamorphic telescope **605**. Anamorphic telescope **605** converts the elongated beam to a circular beam for a symmetrical focus for the coupling optics **516**. The optical signals are then passed from the anamorphic telescope **605** to the coupling optics **516** where they are coupled to the output ports **3114**.

[0227] The anamorphic telescopes described herein may have unit magnification in the dispersion plane. The anamorphic telescope **603** and/or **605** may have a non-unitary magnification in the dispersion axis. This enables control of the separation of the ports, which aids fabrication.

[0228] The structure described with reference to FIGS. **31** and **28a** and **28b** may be used as an add-drop part of an edge ROADM of the type described with reference to FIG. **19**.

[0229] Many of the examples described herein refer to an adWSS comprising spectra on one programmable deflection plane and space switches on another programmable deflection plane.

AdWSSs are less complex and cheaper to manufacture than $M \times N$ WSSs, and hence preferred where the output spectrum is coarse and there are a large number of output ports each of which takes only a small number of input channels. However, drop ports can only take channel data from one input port at once. Each of the example adWSS structures described herein could instead be $M \times N$ WSSs. In these cases, the beam steering optical device **511** remaps the de-multiplexed spectra incident on it so as to rearrange the geometric arrangement of the channels to match the geometric arrangement of the output ports. The beam steering optical device **511** outputs the remapped de-multiplexed spectra to the second programmable deflection plane. This enables full channel control.

[0230] According to other examples, an adWSS may comprise space switches on two programmable deflection planes. FIGS. **29**, **30** and **31** illustrate examples of adWSSs in which the arrangement of the input ports is different to that of the output ports and which comprise space switches on two programmable deflection planes. The features of FIGS. **29**, **30** and **31** which are the same as those of FIG. **5** and operate in the same manner are illustrated with the same reference numerals. The detail of those features is as described above with respect to FIGS. **5**, **6a** and **6b**.

[0231] FIG. **29** illustrates an adWSS **2900** which operates in a similar way to that shown in FIG. **11**. The structure and operation of the adWSS of FIG. **29** differs from the description of the adWSS of FIG. **11** by the following details.

[0232] In FIG. **29**, the input ports **2901** are arranged in a column array and the output ports **2902** are arranged in a grid array. The coupling lenses **503** match the distribution of the input ports **2901**. The coupling lenses **516** matches the distribution of the output ports **2902**. In contrast to the adWSS of FIG. **11**, there is no diffraction grating positioned between lenses **504** and **506**. Instead, input beams pass through coupling lenses **503**, lens **504** and lens **506** before being incident on the first programmable deflection plane **508**. Multiplexed beams are therefore incident on the first programmable deflection plane **508**. The array of space switches **2903** on the first programmable deflection plane **508** is in the same conformal shape as the arrangement of the input ports, i.e. a column array in this example. The array of beams from the first programmable deflection plane **508** travels through a lens **2904**. Lens **2904** may be a Fourier transform lens. The lens **2904** may form a conjugate plane of the deflected number of beams on the entrance of the beam steering optical device **511**. The beam steering optical device **511** remaps the deflected array of beams such that the remapped array of beams directed towards the second programmable deflection plane **514** matches the conformal shape of the output ports **2902**. In this case, the shape is a 2×3 rectangular grid. The space switches **2905** on the second programmable deflection plane **514** match the conformal shape of the output ports. The arrangement of the space switches **2905** matches the conformal shape of the remapped array of beams. Thus, the optical device **511** remaps the vertical column of deflected beams input to it to a two-dimensional array of beams for imaging onto the space switches **2905** via the telescopic lenses **512** and **513**. The space switches **2905** then deflect the remapped array of beams to the output ports **2902** via lens **2906**, lens **2907** and coupling lenses **516**. Lenses **2906** and **2907** may form a 4F optical system.

[0233] FIG. **30** illustrates the same adWSS **2900** seen in FIG. **29**. FIG. **30** shows that the adWSS **2900** can work in reverse. In other words, the adWSS **2900** is fundamentally directionless. FIG. **30** shows that when the adWSS **2900** is operated in reverse, the ports **2902** can be used as input ports and the ports **2901** act as output ports. In other words, input beams are directed from ports **2902** to ports **2901**.

[0234] From input ports **2902**, input beams pass through coupling lenses **516**, lens **2907** and lens **2906**. Input beams are then incident on the second programmable deflection plane **514**. The array of space switches **2905** on the second programmable deflection plane **514** is in the same conformal shape as the arrangement of the input ports, i.e. a 2×3 grid in this example. The array of beams from the second programmable deflection plane **514** travels through lenses **513** and **512** to the beam steering optical device **511**. The beam steering optical device **511** remaps the deflected array

of beams such that the remapped array of beams directed towards the first programmable deflection plane **508** matches the conformal shape of the output ports **2901**. In this case, the shape is a column array. The remapped array of beams pass through lens **2904**. Lens **2904** may be a Fourier transform lens. The lens **2904** may form a conjugate plane of the deflected number of beams on the first programmable deflection plane **508**. The space switches **2903** on the first programmable deflection plane **508** match the conformal shape of the output ports **2901**. The arrangement of the space switches **2905** matches the conformal shape of the remapped array of beams. Thus, the optical device **511** remaps the two-dimensional array of beams input to it to a vertical column of deflected beams onto the space switches **2903**. The space switches **2903** then deflect the remapped array of beams to the output ports **2901** via lenses **506** and **504** and coupling ports **503**.

[0235] FIG. **31** illustrates an adWSS **3100** which operates in a similar way to that shown in FIG. **13**. The adWSS **3100** of FIG. **31** also includes elements which are common to the adWSS **2900** seen in FIG. **29**. The structure and operation of the adWSS of FIG. **31** differs from the description of the adWSS of FIG. **11** and the adWSS of FIG. **29** by the following details.

[0236] In FIG. **31**, the input ports **3101** are arranged in a 2×3 grid array and the output ports **3102** are arranged in a 3×3 grid array. The coupling lenses **503** match the distribution of the input ports **3101**. The coupling lenses **516** match the distribution of the output ports **3102**. In contrast to the adWSS of FIG. **13**, there is no diffraction grating positioned between lenses **504** and **506**, such that input beams pass through coupling lenses **503**, lens **504** and lens **506** before being incident on the first programmable deflection plane **508**. Multiplexed beams are therefore incident on the first programmable deflection plane **508**. The array of space switches **3104** on the first programmable deflection plane **508** is in the same conformal shape as the arrangement of the input ports, i.e. a 2×3 grid in this example. The array of beams from the first programmable deflection plane **508** travels through lenses **509** and **510** towards the beam steering device. In contrast to the adWSS seen in FIG. **13**, there is no diffraction grating positioned between lenses **509** and **510**. The multiplexed beams from the first programmable deflection plane **508** are therefore sent to the beam steering device **511** without being multiplexed or demultiplexed. As per the example of FIG. **13**, deflected multiplexed light beams are incident on the beam steering device **511**. As explained with respect to FIG. **13**, optical device **511** comprises optical parts **1307** and **1309** and lens **1308**. First, optical part **1307** of optical device **511** converts the double column array of deflected beams from the first programmable deflection plane **508** to a single column array. This single column array is then passed through conjugate Fourier lens **1308** to second optical part **1309**. Second optical part **1309** converts the single column array to a three column array matching the array of space switches **3105** on the second programmable deflection plane **514**. This three column array is then imaged by telescopic lenses **512** and **513** to the space switches **3105** on second programmable deflection plane **514**. The space switches **3105** deflect the remapped array of beams to the output ports **3102** via lens **2906**, lens **2907** and coupling lenses **516**. Lenses **2906** and **2907** may form a 4F optical system.

[0237] FIG. **32** illustrates a further example of a $M \times N$ WSS **3200**. Input light beams from N input ports **3201** pass through a first optical structure **3208** before reaching a beam steering device **511**. When the beams leave the beam steering device **511**, they pass through a second optical structure **3208'** before reaching output ports **3202**. The second optical structure **3208'** may be identical to the first optical structure **3208**, for example when $M=N$. In other examples, the first and second optical structures **3208**, **3208'** may not be identical.

[0238] The first optical structure **3208** comprises a fan lens **3203**, a flat mirror **3204**, an optical system **3205**, a first programmable deflection plane **3206** and a 4F optical system **3207**. The second optical structure comprises a fan lens **3203'**, a flat mirror **3204'**, an optical system **3205'**, a second programmable deflection plane **3206'** and a 4F optical system **3207'**.

[0239] From input ports **3201**, the input light beams pass through the fan lens **3203** which focusses the light through an aperture in the flat mirror **3204**. The flat mirror is a mirror optic which may be referred to as a split aperture mirror (SAM). The input light beams then pass through the optical

system **3205** towards the first programmable deflection plane **3206**. The optical system **3205** may comprise a 4F optical system. The optical system **3205** images the optical signal in the dispersion plane, for example using two 4F dispersion lenses. The optical system may additionally form a Fourier conjugate in a direction normal to the beam path, along the steering axis. For example, the optical system **3205** may further include a lens in the steering plane located between two 4F dispersion lenses. The lens may be a Fourier lens.

[0240] The light is imaged in both directions onto the first programmable deflection plane **3206**. Deflected light is then passed back through the optical system **3205** into the SAM **3204**. As seen in FIG. **34**, when the light passes back through the SAM, it is deflected by the SAM in a direction away from the input ports **3201**. The deflected light then passes through 4F optical system **3207** and is incident on the beam steering device **511** of the type seen in FIGS. **13** and **31**. The beam steering device **511** comprises optical parts **1307** and **1309** and lens **1308**. The lens **1308** may be a conjugate Fourier lens.

[0241] From the beam steering device **511**, the light is passed in through the second optical structure **3208'** which may be identical to the first optical structure **3208**, towards the output ports **3202**. In other words, the light beams pass through the 4F optical system **3207'**, the SAM **3204'** and the optical system **3205'** before being deflected by the second programmable deflection plane **3206'** back through the optical system **3205'**, SAM **3207'** and lens **3203'** towards the output ports **3202**. The arrangement shown in FIG. **32** has the advantage that the input ports **3201** and the output ports **3202** may be separated laterally by a small distance. Despite the small distance between the input and output ports, the described arrangement reduces crosstalk between the two sets of ports. The input ports **3201** and output ports **3202** may be positioned parallel to one another as seen in FIG. **32**.

[0242] The mirror optics (SAMs) **3204** and **3204'** seen in FIG. **32** may take a variety of forms, as illustrated in FIGS. **33(a)** to **(e)**. In other words, there are a number of different types of SAM which could be used in the switch **3200**. FIG. **33** illustrates five possible types of SAM (**3302**, **3303**, **3305**, **3306**, **3307**) but other types may exist.

[0243] Each FIG. **33(a)** to **(e)** illustrates an optical structure to which light **3301** is input. In FIGS. **33(a)** and **33(b)**, the input light is not parallel to the steering axis. The input light may be known as “off-axis” light. In FIGS. **33(c)** to **(e)**, the input light is parallel to the steering axis. This may be known as “on-axis” light. The structure to which the light in each case is input includes a fan lens **3203**, an optical system **3205** and a programmable deflection plane **3206**, as previously described with respect to FIG. **32**. As previously described, the optical system **3205** may comprise a set of two 4F lenses configured to image the signal in the dispersion plane and a lens configured to form a Fourier conjugate in the steering direction. FIGS. **33** illustrate only the set of 4F dispersion lenses. Any components of the optical system **3205** in the steering plane are not shown.

[0244] In FIG. **33(a)**, the SAM **3302** takes the form of an edge of a mirror located near the input light beam. This type of SAM may be referred to as a “knife edge”. FIG. **33(a)** shows that due to the direction of the input light and position of the mirror **3302**, the input light passes by the edge of the mirror. In other words, the input light is not deflected by the SAM **3302**. Once the light has been deflected by the programmable deflection plane **3206** and has passed back through optical system **3206**, the deflected light is intercepted by the mirror **3302** and deflected such that the light is directed away from the input light. As illustrated in FIG. **33(a)**, the deflected light is not intercepted by the mirror **3302** at its focus. Instead, the mirror intercepts the light ahead of its focus.

[0245] FIG. **33(b)** illustrates the same arrangement as that seen in FIG. **33(a)** with the addition of a deflection optic **3304** located between the two 4F dispersion lenses in the optical system **3205**. The dispersion optic changes the angle of the deflected light in the Fourier plane between the two 4F lenses so as to create a positional change in the light. The result of this positional change is that the mirror **3303** intercepts the deflected light at its focus, as shown in FIG. **33(b)**.

[0246] In FIG. **33(c)**, the SAM **3305** is a traditional split aperture mirror. The SAM **3305** comprises

two mirrors separated by a gap. In the example shown, the two mirrors are displaced from one another in the steering direction. FIG. 33(c) shows that the input light is focused by the fan lens **3203** so that it passes through the gap between the two mirrors. In other words, the input light passes by the SAM **3305** without being deflected. Once the light has been deflected by the programmable deflection plane **3206** and has passed back through optical system **3205**, the deflected light is intercepted by one of the two mirrors of the SAM **3305** at its focus. As mentioned above, the optical system **3205** includes a steering lens (not shown), which may be positioned between the two 4F dispersion lenses. The steering lens forms a Fourier conjugate focus in the steering direction at the SAM mirror plane such that the light is intercepted by one of the two mirrors of the SAM **3305**. In other words, the steering lens of the optical system **3205** deflects the light deflected by the programmable deflection plane **3206** in the steering direction so that instead of passing through the gap between the two mirrors, the light is incident on one of the two mirrors forming the SAM **3305**. The light is then deflected by the intercepting mirror such that the light is directed away from the input light.

[0247] In FIG. 33(d), the SAM **3306** is a device which remaps the beams incident on it to new locations. The SAM **3306** may be the beam steering device **511** previously described. The SAM **3306** comprises a mirror array and has a form similar to the example beam steering device seen in FIG. 16. The example SAM **3306** seen in FIG. 33(d) comprises a first set of two mirrors **3306a**, **3306b** separated by a gap and a second set of mirrors **3306c**, **3306d** spaced apart from the first set of mirrors, the second set of mirrors being positioned side-by-side without a gap between them. In this example, the first set of two mirrors **3306a**, **3306b** are displaced from one another in the steering direction, with a gap between them along the steering axis. The second set of mirrors **3306a**, **3306b** are displaced from one another in the dispersion direction, without a gap between them along the dispersion axis. As previously explained with respect to FIG. 16, each group of points incident on the device reflects off a pair of mirror surfaces and follows a different optical path through the mirror array. According to further examples, the mirror array of the SAM **3306** may have a different configuration to that shown in FIG. 33(d).

[0248] FIG. 33(d) shows that input light is focused by the fan lens **3203** so that it passes through the gap between the two mirrors of the first set of mirrors **3306a**, **3306b**. In other words, the input light passes by the SAM **3306** without being deflected. Once the light has been deflected by the programmable deflection plane **3206** and has passed back through optical system **3205**, the deflected light is intercepted by one of mirrors **3306a**, **3306b** of the SAM **3306**. The light is reflected by one of the mirrors **3306a**, **3306b** onto one of mirrors **3306c**, **3306d**. FIG. 33(d) shows a group of points which are reflected from mirror **3306a** onto mirror **3306c**, but other groups of points may take other paths through the device. The reflected light is further reflected by mirror **3306c**. The light is thus deflected by the SAM **3306** such that it is directed away from the input light.

[0249] In FIG. 33(e), the SAM **3307** is a device which alters the configuration of beams incident on it so as to generate a gap between a first set of those beams and a second set of those beams. The SAM **3307** may be the optical structure **2502** previously described. The SAM **3307** comprises a mirror array and has a form similar to the example optical structure seen in FIG. 25. The example SAM **3307** seen in FIG. 33(e) comprises a first set of two mirrors **3307a**, **3307b** separated by a gap and a second set of mirrors **3307c**, **3307d** spaced apart from the first set of mirrors, the second set of mirrors being positioned without a gap between them. In this example, the first set of two mirrors **3307a**, **3307b** are displaced from one another in the steering direction, with a gap between them along the steering axis. The second set of mirrors **3307a**, **3307b** are displaced from one another in the steering direction, without a gap between them along the steering axis. As previously explained with respect to FIG. 25, a portion of optical signals incident on the device are incident on the mirror **3307a** and a portion is incident on the mirror **3307b**.

[0250] FIG. 33(e) shows that input light is focused by the fan lens **3203** so that it passes through

the gap between mirrors **3307a** and **3307b**. In other words, the input light passes by the SAM **3307** without being deflected. Once the light has been deflected by the programmable deflection plane **3204** and has passed back through optical system **3205**, the deflected light is intercepted by one of mirrors **3307a**, **3307b** of the SAM **3307**. The light is reflected by one of the mirrors **3307a**, **3307b** onto one of mirrors **3307c**, **3307d**. FIG. **33(e)** shows a portion of the light which is reflected from mirror **3307a** onto mirror **3307c**. Other portions of light will be reflected from mirror **3307b** to mirror **3307d**. The reflected light is further reflected by mirror **3307c**. The light is thus deflected by the SAM **3307** such that it is directed away from the input light and such that the light is split into two separate sets of beams.

[0251] FIG. **34** illustrates an arrangement **3400** of multiple of the WSSs **3200** seen in FIG. **32**. In the arrangement **3400** there are K first optical structures **3208**. In the example seen in FIG. **34**, K=6 (i.e. the arrangement includes first optical structures **3208(a)** to **(f)**). The arrangement **3400** further includes P second optical structures **3208'**. In the example seen in FIG. **34**, P=6 (i.e. the arrangement includes second optical structures **3208'(a)** to **(f)**). In the example seen in FIG. **34**, the arrangement includes the same number of first optical structures **3208** as second optical structures **3208'** i.e. K=P. The illustrated arrangement is therefore symmetrical. According to other examples, the arrangement may not be symmetrical such that $K \neq P$.

[0252] The first and second optical structures **3208**, **3208'** take the forms previously described with respect to FIGS. **32** and **33**. Each first optical structure **3208** therefore includes a first programmable deflection plane **3206**, which may be known as an input plane. Each second optical structure **3208'** includes a second programmable deflection plane **3206'**, which may be known as an output plane. Each first optical structure **3208** includes a SAM **3204**. Each second optical structure includes a SAM **3204'**.

[0253] Each of the first optical structures **3208(a)** to **(f)** includes a set of L input ports. The arrangement **3400** thus includes K sets of L input ports. The total number of input ports present in the arrangement **3400** is therefore equal to $K \times L$. Each of the second optical structures **3208'(a)** to **(f)** includes set of Q output ports. The arrangement **3400** thus includes P sets of Q output ports. The total number of output ports present in the arrangement **3400** is therefore equal to $P \times Q$. In the arrangement **3400**, each pair of optical structures **3208**, **3208'** comprises the same number of input ports as output ports i.e. $L=Q$. In other words, the structure is symmetrical. In other examples, the structure may not be symmetrical. For a pair of optical structures **3208**, **3208'**, the number of input ports L may be different to the number of output ports $L \neq Q$.

[0254] As illustrated, the first and second optical structures **3208** and **3208'** respectively direct light to and from a common beam steering device **511**. In other words, light from each of N input ports is channelled through the same beam steering device **511**. The beam steering device **511** takes the same form as that seen in FIG. **32**. As previously described and as illustrated in FIG. **34**, for each of the first optical structures **3208(a)** to **(f)**, deflected light deflected by the SAM **3204** in the first optical structure **3208** is directed to the beam steering device **511** and is incident on the first optical part **1307** of the beam steering device **511** at L ports (where L is the number of input ports associated with each first optical structure **3208**). Thus, in the whole arrangement **3400** of FIG. **34**, light from the K sets of L input ports is directed to N ports on the first optical part **1307** of the beam steering device **511**, where $N=K \times L$. The array of L input ports at each input position (i.e. on each of the K first programmable deflection planes **3206**) can be in an arbitrary distribution or array shape and is transposed to a column of N ports by the first optical part **1307**.

[0255] As the N ports pass through the lens **1308**, as previously described, the column of N ports is mapped onto a column of M ports. At the second optical part **1309**, the column of M ports is transposed to a grid of M ports having a different distribution. As previously described, the arrangement **3400** includes P second optical structures **3208'** and each of the second optical structures **3208'(a)** to **(f)** includes set of Q output ports. The arrangement **3400** thus includes P sets of Q output ports. The total number of output ports present in the arrangement **3400** is therefore

equal to $P \times Q$. The grid of M ports are distributed back to the P sets of Q output ports through the second optical structures **3208'**(a) to (f). It is therefore the case that $M = P \times Q$.

[0256] In this example, the first programmable deflection planes (**3206**) and the second programmable deflection planes (**3206'**) are arrayed interleaved with one another. As previously described, the first and second programmable deflection planes may be incorporated onto a single SLM device or may be located on separate SLM devices. The advantage of the arrangement **3400** is that a large number of separate SLMs can be used to create a much larger switch port number N and/or M that would be obtainable from a single SLM, while still utilising only 1D SLM steering. In other words, the number of programmable deflection planes required for a given number of N input ports is reduced such that for a given number of programmable deflection planes (for example a given number of SLM devices), a larger number of N input ports (and/or M output ports) can be used.

[0257] FIG. **35** illustrates a further implementation of the beam steering device **511**. Similar to the beam steering optical device seen in FIG. **16**, the device shown in FIG. **35** comprises a mirror array **3500**. As with the example seen in FIG. **16**, the mirror array **3500** remaps an array of points **3501** incident on it to a grid of points **3504** having a different conformal pattern, as previously described. The beam steering device **511** of FIG. **35** is designed so as to utilise incoming astigmatic light to optimise transmission through the beam steering device. As explained in more detail below, the beam steering device comprises pairs of parallel mirror surfaces.

[0258] FIG. **35** illustrates that the mirror array **3500** includes a first set of four mirror surfaces **3504a**, **3504b**, **3504c**, **3504d** arranged in a column. The mirror array further includes a second set of four mirror surfaces **3605a**, **3605b**, **3605c**, **3605d** arranged in a 2×2 grid. The mirror array **3500** remaps a single column of points incident on it to a grid of points. The mirror array **3500** splits the column points into groups. Each group of points follows a different optical path through the mirror array **3500**. FIG. **35** illustrates just one group **3501** however it will be understood that more groups may be used. Each group of points reflects off a pair of mirror surfaces, the pair of mirror surfaces being parallel to one another.

[0259] The group **3501** includes light to which an astigmatism has been applied. For example, in the $M \times N$ WSS **3200** seen in FIG. **32**, an astigmatism may be applied to the input light by the optical elements positioned between the first programmable deflection plane **3206** and the beam steering device **511**, for example by altering the position and focal length of the steering lenses with respect to the dispersion lenses in the optical system **3205**. Astigmatic light features a difference in focus position along the optical axis for two orthogonal planes. In the example seen in FIG. **35**, these are the orthogonal steering and dispersion planes. As seen in FIG. **35**, the astigmatic light beam **3501** is divided into a portion of light aligned with the dispersion direction **3502** (the dispersion plane) and a portion of light aligned with the steering direction **3503** (the steering plane). In FIG. **35**, the dispersion plane is illustrated by only a pair of lines **3501a** and **3501b** for illustrative purposes. In FIG. **35**, the steering plane is illustrated by only a pair of lines **3501c** and **3501d** for illustrative purposes.

[0260] As seen in FIG. **35**, the input light beam **3501** is incident on first mirror surface **3504b** and second mirror surface **3505b**. Normally, astigmatic light is undesirable but as explained below, in the following example, astigmatic light is used to optimise transmission and minimise losses through the mirror array. It is not necessary in the beam steering device for both the steering plane and the dispersion plane of the incoming astigmatic light **3501** to be focused by a single mirror surface. It is however important that when the light beam is incident on the vertical column of mirror surfaces **3504a-d**, that the beam is in focus along the steering axis such that the beam is incident on only one of mirrors **3504a-d**. It is not important at this stage whether or not the beam is focused along the dispersion axis. In contrast, when the light beam is incident on the 2×2 grid of mirror surfaces **3505a-d** which includes mirrors spaced along the dispersion axis, it is important that the beam is in focus in the dispersion direction. As shown in FIG. **35**, the steering plane **3501c**,

3501d is focused on first mirror surface **3504b**. The dispersion plane **3501a**, **3501b** is focused on the second mirror surface **3505b**.

[0261] In contrast to the beam steering device seen in FIG. **16** in which each mirror surface of a pair of mirror surfaces are angled with respect to each other, in the example seen in FIG. **35**, each mirror surface in a pair is parallel with respect to the other mirror surface in the pair. Specifically, in this example, mirror surface **3504a** is parallel to mirror surface **3505a**, mirror surface **3504b** is parallel to mirror surface **3505b**, mirror surface **3504c** is parallel to mirror surface **3505c** and mirror surface **3505d** is parallel to mirror surface **3505d**.

[0262] When incoming light is reflected sequentially by two mirror surfaces which are not parallel, the light input to the beam steering device (and therefore its associated optical image) is rotated about the optical axis. In other words, the direction of light entering the beam steering device is not the same direction as the light leaving the device. Such rotation can be accounted for by optical components later in the path taken by the incoming light, but it is generally preferable to eliminate any rotation of the optical image input to the device. The beam steering devices seen in FIGS. **35** and **36** therefore incorporate pairs of parallel mirror surfaces into the first and second planes of mirror surfaces. The result of incoming light being incident on pairs of parallel mirror surfaces is that the direction of light entering the beam steering device is the same direction as the light leaving the device and no rotation of the optical image about the optical axis occurs. There is therefore no need for additional optical components in the system to later correct for such rotation. An analogous approach can be taken with regards to the optical structures described herein configured to alter the configuration of incident beams so as to generate a gap between those beams.

[0263] It is undesirable to input astigmatic light to the second programmable deflection plane or the output ports. The beam steering device **511** seen in FIG. **35** may therefore be used as part of a switch having an optical system configured to correct out the astigmatism. For example, in the $M \times N$ WSS **3200** seen in FIG. **32**, an optical system configured to correct out the astigmatism may be positioned between the beam steering device **511** and the second programmable deflection plane **3206'**. Any of the previously described anamorphic telescopes may be modified so as to correct for the astigmatism in addition to making the input light beam more circular, as described. In the same way as the astigmatism is applied to the light by the optical system **3205**, the position and focal length of the steering lenses with respect to the dispersion lenses in the optical system **3205'** may be adjusted so as to remove the astigmatism after the light has passed through the mirror array **3500**.

[0264] FIGS. **36(a)**, **36(b)** and **36(c)** illustrate three possible configurations for the mirror surfaces in the beam steering device **511** of the type seen in FIG. **35** comprising pairs of parallel mirror surfaces. FIGS. **36(a)**-(**c**) show examples of mirror arrays in beam steering devices having a view into the dispersion plane (i.e. along the steering axis).

[0265] FIG. **36(a)** illustrates a beam steering device **3600** in which a first plane of mirror surfaces **3602a**, **3602b** are in the form of a column (i.e. they are displaced along the steering axis). A second plane of mirror surfaces **3603a**, **3603b** are laterally spaced i.e. displaced in the dispersion direction. FIG. **36(a)** shows two mirror surfaces in each set of mirror surfaces. It will be understood that each set of mirror surfaces may include more than 2 mirror surfaces, for example 4 mirror surfaces as shown in FIG. **35**. As seen in FIG. **36(a)**, the light (**3601a**) which is incident on mirror surface **3602a** is reflected onto and then reflected by mirror surface **3603a**. The light (**3601b**) which is incident on mirror surface **3602b** is reflected onto and then reflected by mirror surface **3603b**. Mirror surfaces **3602a** and **3603a** are parallel to one another. The light incident on the first plane of mirror surfaces **3602a**, **3602b** is therefore parallel to light leaving the second plane of mirror surfaces **3603a**, **3603b**. As previously explained with respect to other beam steering devices, it is important that that other conditions are observed, such as the fact that the path length through the mirror array should be the same for all beam paths so that a coincident focus can be achieved for all

beams.

[0266] The mirror array of FIG. **36(a)** further includes a flat directing mirror **3604**. In the example seen in FIG. **36(a)**, the mirror is used to change the direction of light leaving the second plane of mirror surfaces by 90 degrees. The angle between the light **3301** entering the beam steering device **3600** and the light leaving the device is therefore 90 degrees. The plane of the mirror **1604** is normal to the dispersion plane. To avoid rotation of the optical signal as previously discussed, the mirror **3604** may be rotated relative to mirror surfaces **3602**, **3603** about any axis normal to the optical axis direction.

[0267] FIG. **36(b)** illustrates a further beam steering device **3600'** which includes all the features described as being part of the device **3600**. In addition, the mirror array includes a further flat directing mirror **3605**. The angle between the light **3601** entering the beam steering device **3600'** and the light leaving the device is therefore 180 degrees. FIG. **36(b)** therefore illustrates that multiple flat directing mirrors can be used to direct the beam leaving the second plane of mirror surfaces. The flat directing mirrors **3604**, **3605** do not cause a rotation of light incident on them about the optical axis as they are positioned such that the plane of each of the mirrors is normal to the optical axis (and the dispersion axis). Other beam steering devices may include a different number of flat directing mirrors. As previously described, any of the mirror surfaces described may be replaced with any reflecting surface, for example a prism exhibiting total internal reflection, multilayer interference surfaces or polarisation-dependent beam-splitting elements.

[0268] FIG. **36(c)** illustrates a further beam steering device **3606** featuring the same first plane of mirror surfaces **3602a**, **3320b** and second plane of mirror surfaces **3603a**, **3603b**. An optical system **3607** is positioned between the first and second planes of mirror surfaces. FIG. **36(c)** illustrates an example in which the optical system **3607** is a cylindrical Fourier lens aligned with the dispersion axis. According to other examples, the optical system **3607** may comprise one or more mirrors, diffractive and/or holographic elements. The optical system **3607** may comprise a sequence of normal and/or freeform lenses. According to further examples, multiple lenses may be positioned between the two planes of mirror surfaces **3602**, **3603**. For example, a 4F optical system may be positioned between the first and second planes of mirror surfaces. The optical system **3607** may comprise any element with optical power. The optical system is used to image the light leaving the first plane of mirror surfaces onto the second plane of mirror surfaces. In general, the optical system **3607** may be any known imaging system which would enable the beam steering device **3606** to remap the incoming array of beams **3601** as previously described.

[0269] For the same reasons, it can be advantageous to use pairs of parallel mirror surfaces in optical structures such as those shown in FIG. **25**. FIGS. **37(a)**, **37(b)** and **37(c)** illustrate further examples of optical structures configured to alter the configuration of the beams incident on it so as to generate a gap between those beams, the structures comprising pairs of parallel mirrors.

[0270] FIG. **37(a)** illustrates an optical structure **3700** comprising a mirror array which comprises a first mirror assembly **3702** and a second mirror assembly **3703**. As per the optical structure illustrated in FIG. **25**, each mirror assembly comprises two mirror surfaces. The first mirror assembly **3702** comprises a first mirror surface **3702a** and a second mirror surface **3702b**. The second mirror assembly **3703** comprises a third mirror surface **3703a** and a fourth mirror surface **3703b**. The first mirror surface **3702a** and the second mirror surface **3702b** are displaced relative to one another along the steering axis. The third mirror surface **3703a** and the second mirror surface **3703b** are displaced relative to one another along the steering axis. The viewpoint of FIG. **37** is into the dispersion plane (i.e. along the steering axis).

[0271] As previously described with respect to FIG. **25**, optical signals **3701** entering the mirror array are incident on the first mirror assembly **3702**. A portion of the optical signals entering the mirror array are incident on the first mirror surface **3702a**. A portion of the optical signal **3701** entering the mirror array are incident on the second mirror surface **3702b**. The first and second mirror surfaces are angled away from each other. Thus, when parallel optical signals are incident

on the first and second mirror surfaces, the first group of optical signals reflected by the first mirror surface diverge from the second group of optical signals reflected by the second mirror surface. A gap thereby forms between the divergent first and second groups of optical signals. These non-parallel divergent first and second groups of optical signals are then routed to the second mirror assembly **3703**. The light reflected by the first mirror surface **3702a** (the first group of optical signals) is directed to the third mirror surface **3703a** of the second mirror assembly **3703**. The light reflected by the second mirror surface **3702b** (the second group of optical signals) is directed to the fourth mirror surface **3703b** of the second mirror assembly **3703**. Third mirror surface **3703a** reflects this first group of optical signals out of the mirror array. Fourth mirror **3703b** reflects this second group of optical signals out of the mirror array in a direction parallel to the first group of optical signals. The third mirror **3703a** and fourth mirror **3703b** are inverse to each other. The first and second group of optical signals output from the mirror array **3700** are parallel and separated from each other by a gap. Mirror surfaces **3702a** and **3703a** are parallel to one another. The portion of the optical signals incident on the first mirror surface **3702a** is therefore parallel to the portion of the optical signal leaving the third mirror surface **3703a**. Mirror surfaces **3702b** and **3703b** are parallel to one another. The portion of the optical signals incident on the first mirror surface **3702b** is therefore parallel to the portion of the optical signal leaving the third mirror surface **3703b**. As previously described, any rotation of the optical image input to the mirror array is therefore eliminated. The mirror array of FIG. **37(a)** further includes a flat directing mirror **3704**, which takes the same form as mirror **3604** described with respect to FIG. **36(a)**.

[0272] FIG. **37(b)** illustrates a further optical device **3700'** which includes all the features described as being part of the device **3700**. In addition, the device includes a further flat directing mirror **3705**. The angle between the light **3701** entering the optical device **3700'** and the light leaving the device is therefore **180** degrees. FIG. **37(b)** therefore illustrates that multiple flat directing mirrors can be used to direct the beam leaving the second array of mirror surfaces **3703**. The flat directing mirrors **3704**, **3705** do not cause a rotation of light incident on them about the optical axis as they are positioned such that the plane of each of the mirrors is normal to the optical axis. Other beam steering devices may include a different number of flat directing mirrors. As previously described, any of the mirror surfaces described may be replaced with any reflecting surface, for example a prism exhibiting total internal reflection, multilayer interference surfaces or polarisation-dependent beam-splitting elements.

[0273] FIG. **37(c)** illustrates a further optical device **3706** featuring the same first array of mirror surfaces **3702** second array of mirror surfaces **3703**. An optical system **3707** is positioned between the first and second planes of mirror surfaces. The optical system **3707** may comprise one or more mirrors, diffractive and/or holographic elements. The optical system **3707** may comprise a sequence of normal and/or freeform lenses. According to further examples, multiple lenses may be positioned between the two planes of mirror surfaces **3702**, **3703**. For example, a 4F optical system may be positioned between the first and second planes of mirror surfaces. The optical system **3707** may comprise any element with optical power. The optical system is used to image the light leaving the first plane of mirror surfaces onto the second plane of mirror surfaces. In general, the optical system **3707** may be any known imaging system which would enable the optical device **3706** to create a spatial gap in the incoming array of beams **3701** as previously described.

[0274] FIG. **38** illustrates a further optical device **3800** of the type configured to alter the configuration of the beams incident on it so as to generate a gap between the those beams. The device includes a mirror array comprising a first array of mirror surfaces **3802** and a second array of mirror surfaces **3803**. As previously described, the mirror array comprises pairs of parallel mirror surfaces. The mirror surface **3802a** of the first array of mirror surfaces seen in FIG. **38** is parallel to the mirror surface **3803a** of the second array of mirror surfaces. The array further includes three flat directing mirrors **3804**, **3805**, **3806**. The device **3800** may be utilised in a WSS such as the M×N WSS **3200** seen in FIG. **32**.

[0275] The optical device **3800** may be positioned as part of the first optical structure **3208** or the second optical structure **3208'**. The optical device **3800** may replace SAM **3204** and/or SAM **3204'**, as illustrated in FIG. **33(e)**.

[0276] When the optical device **3800** replaces SAM **3204**, the light input **3801a** to the optical device **3800** may come from the fan lens **3203**. The device **3800** may therefore create a gap in the steering direction between two portions of light **3801b** incident on the optical system **3205**

[0277] In other words, the device **3800** creates a gap in the light's spatial distribution. As previously mentioned, the optical system **3205** may comprise a Fourier lens. The optical system **3205** therefore forms a Fourier conjugate of the incoming light **3205'** in the steering direction such that the light incident on the first programmable deflection plane **3206** has a gap in its angular distribution. The first programmable deflection plane **3206** deflects the incoming light so that the deflected light **3801c** passes through the gap created in light **3801b**. The deflected light **3801c** therefore passes through the mirror array of device **3800** and is reflected by flat mirror **3805**. The device **3800** directs the output light to the 4F optical system **3207** (and onward towards the beam steering device **511**).

[0278] Alternatively, when the optical device replaces SAM **3204'**, as shown in FIG. **38**, the light input **3801a** to the optical device **3800** comes from the 4F optical system **3207'**. The device **3800** may therefore create a gap between two portions of light **3801b** incident on the optical system **3205'** and the second programmable deflection plane **3206'**. The second programmable deflection plane **3206'** directs the incoming light **3801b** so that the deflected light **3801c** passes through the gap created. The deflected light **3801c** therefore passes through the mirror array of device **3800** and is reflected by flat mirror **3805**. The device directs the output light to the fan lens **3203'** (and onward towards the output pots **3202**). The device may also be used for light travelling in the opposite directions to those shown and described i.e. the device is reversible.

[0279] Suitably, the programmable deflection planes described herein are SLM planes. The SLM plane may be a MEMS mirror array. The SLM plane may be an LCOS device. The LCOS device applies a hologram to enable a beam deflection. Alternatively, the SLM plane may be another locally configuration device capable of applying a deflection to a channel from the input.

[0280] In the examples described herein the first and second programmable deflection planes are incorporated onto a single SLM device. This utilises fewer components, reduces control complexity and hence cost compared to if the programmable deflection planes are on separate SLM devices. Alternatively, the programmable deflection planes may be on separate SLM devices.

[0281] In the examples described herein, optical components are used to route data through the described switches. There is no absorption and re-emission of light, thereby avoiding the lag associated with transmitting data through electronical switches. The optical components also have lower power consumption than equivalent electronical implementations.

[0282] The beam steering optical devices described herein comprise prism and/or mirror structures. These enable more efficient use of the SLM panel area thereby allowing significantly improved port count performance for a given SLM resolution. However, the beam steering optical devices could be implemented using other optical elements which achieve the same effect. For example, single or multiple holographic or diffractive optical elements or further active SLM planes may be used to achieve the same optical effect.

[0283] The examples described herein incorporate a diffraction grating. However, any demultiplexer which demultiplexes light signals into spatially separated data channels may be used instead of a diffraction grating in any of the examples. Similarly, any multiplexer which multiplexes spatially separated data channels into multiplexed light signals may be used instead of a diffraction grating in any of the examples. Any suitable optical dispersion device may be used as a multiplexer and/or a demultiplexer.

[0284] Lenses and other optical components described herein as single structures may be implemented using assemblies having a plurality of components which achieve the same optical

effect. Examples of such assemblies are: achromatic doublets, achromatic triplets, Cook doublets, telescopes and microscopes for imaging lenses. The lenses described herein may be implemented using other optics with the same optical power. For example, a curved mirror may be used as a lens. For example, multiple lens elements, mirrors, mirror arrays, catadioptric systems, holographic optical elements or diffractive optical elements may be used as a lens. Separate elements, for example 4F lenses, with the same optical properties can be arranged as separate passes through or from a single physical element.

[0285] The applicant hereby discloses in isolation each individual feature described herein and any combination of two or more such features, to the extent that such features or combinations are capable of being carried out based on the present specification as a whole in the light of the common general knowledge of a person skilled in the art, irrespective of whether such features or combinations of features solve any problems disclosed herein, and without limitation to the scope of the claims. The applicant indicates that aspects of the present invention may consist of any such individual feature or combination of features. In view of the foregoing description it will be evident to a person skilled in the art that various modifications may be made within the scope of the invention.

Claims

1. An optical switch comprising: a set of input ports, each input port configured to transport an optical signal having at least one component frequency channel; a set of output ports, each output port configured to transport an optical signal having at least one component frequency channel; a first programmable deflection plane configured to deflect beams incident on it from the set of input ports to form a deflected array of beams; a second programmable deflection plane configured to deflect beams incident on it towards the set of output ports; and a beam steering optical device positioned in the optical path between the first and second programmable deflection planes, the beam steering optical device configured to remap the deflected array of beams from the first programmable deflection plane to form a remapped array of beams directed to the second programmable deflection plane, by changing a spatial positioning and/or orientation of each beam from the deflected array of beams in the remapped array of beams such that the spatial positioning and/or orientation of at least one beam of the deflected array of beams is changed differently to at least one other beam of the deflected array of beams, whereby the beam steering optical device is configured to split the deflected array of beams into groups of beams, where each group of beams is incident on a different set of optical elements to another group of beams thereby causing each group of beams to follow a different optical path through the beam steering optical device to the other groups of beams so as to cause the geometrical distribution of the groups of beams to change.
2. An optical switch as claimed in claim 1, wherein the beam steering optical device is configured to remap the deflected array of beams by causing them to undergo two individual reflections, refractions or diffractions.
3. An optical switch as claimed in claim 1, wherein the beam steering optical device is configured to remap the deflected array of beams by transposing them in two orthogonal directions.
4. An optical switch as claimed in claim 1, wherein the deflected array of beams has a linear distribution, and the remapped array of beams has a non-linear distribution.
5. An optical switch as claimed in claim 1, wherein the deflected array of beams is a single column array, and the remapped array of beams is a two-dimensional array or the deflected array of beams is a two-dimensional array, and the remapped array of beams is a single column array.
6. (canceled)
7. An optical switch as claimed in claim 1, wherein the deflected array of beams is a two-dimensional array having a first configuration, and the remapped array of beams is a two-dimensional array having a different configuration to the first configuration.

- 8.** An optical switch as claimed in claim 1, wherein the beam steering optical device comprises one or more of the following passive components: a prism, a mirror array, a catadioptric system, a freeform lens and mirror, a diffractive optical element, a diffractive optical element array, a holographic optical element and a holographic optical element array.
- 9.** An optical switch as claimed in claim 1, wherein the beam steering optical device comprises one or more of the following electrically switchable components: an optical microelectromechanical system (MEMS) array, and a liquid crystal on silicon (LCOS) device.
- 10.** An optical switch as claimed in claim 1, wherein the beam steering optical device comprises a first optical element, a second optical element and a conjugate Fourier lens, wherein the conjugate Fourier lens is positioned in the optical path between the first and second optical elements.
- 11.** An optical switch as claimed in claim 1, wherein the beam steering optical device comprises a prism comprising: a first retroreflecting structure configured to provide an optical path for a first set of beams of the deflected array of beams; and a second retroreflecting structure configured to provide an optical path for a second set of beams of the deflected array of beams through a periscope prism, so as to alter the conformal arrangement of the first and second sets of beams of the deflected array of beams thereby forming the remapped array of beams having a geometrical distribution which matches the geometrical distribution of the image of the set of output ports on the second programmable deflection plane.
- 12.** An optical switch as claimed in claim 1, wherein the beam steering optical device comprises a mirror array comprising: a first pair of mirrors configured to provide an optical path for a first set of beams of the deflected array of beams; and a second pair of mirrors configured to provide an optical path for a second set of beams of the deflected array of beams, the first and second pairs of mirrors having differently angled surfaces so as to alter the conformal arrangement of the first and second sets of beams of the deflected array of beams thereby forming the remapped array of beams having a geometrical distribution which matches the geometrical distribution of the image of the set of output ports on the second programmable deflection plane.
- 13.** An optical switch as claimed in claim 1, wherein the beam steering optical device comprises a prism comprising two orthogonal Amici Roof prisms configured to provide an optical path for the deflected array of beams so as to remap the deflected array of beams from a first configuration in a first direction to form the remapped array of beams having the first configuration in a direction orthogonal to the first direction.
- 14.** (canceled)
- 15.** An optical switch as claimed in claim 1, wherein the beam steering optical device comprises three differently angled reflecting surfaces configured to provide an optical path for the deflected array of beams so as to remap the deflected array of beams from a first configuration in a first direction to form the remapped array of beams having the first configuration in a direction orthogonal to the first direction.
- 16.** (canceled)
- 17.** An optical switch as claimed in claim 1, comprising a single spatial light modulator (SLM) panel comprising both the first programmable deflection plane and the second programmable deflection plane.
- 18.** An optical switch as claimed in claim 17, wherein the images of the beams on the first programmable deflection plane and the images of the remapped array of beams on the second programmable deflection plane cover over 80% of the area of the SLM panel.
- 19.** An optical switch as claimed in claim 1, further comprising an anamorphic correction telescope configured to focus the output of each input port of the set of input ports to an anamorphic focus image at an anamorphic focus point; wherein the beam steering optical device comprises: a first structure positioned adjacent to and above the anamorphic focus point, and a second structure positioned adjacent to and below the anamorphic focus point.
- 20.** An optical switch as claimed in claim 1, further comprising a lens configured to focus the

output of each input port of the set of input ports to an input circular focus image at an input circular focus point; wherein the beam steering optical device comprises: a first structure positioned adjacent to and above the input circular focus point, and a second structure positioned adjacent to and below the input circular focus point.

21-29. (canceled)

30. An optical switch as claimed in claim 12, wherein the first pair of mirrors comprises a first mirror and a second mirror, the first and second mirrors being parallel to one another; and the second pair of mirrors comprises a third mirror and a fourth mirror, the third and fourth mirrors being parallel to one another.

31. An optical switch as claimed in claim 30, wherein the first set of beams of the deflected array of beams is incident on the first mirror in a first direction and leaves the second mirror in the first direction; and the second set of beams of the deflected array of beams is incident on the third mirror in a second direction and leaves the fourth mirror in the second direction.

32. (canceled)

33. An optical switch as claimed in claim 1, wherein the optical switch comprises a split aperture mirror positioned in the optical path between the set of input ports and the first programmable deflection plane and between the first programmable deflection plane and the beam steering optical device and/or wherein the optical switch comprises a split aperture mirror positioned in the optical path between the beam steering optical device and the second programmable deflection plane and between the second programmable deflection plane and the set of output ports.

34. (canceled)

35. (canceled)
